

A Macro-finance Model with Crisis Dynamics*

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Abstract

Financial crises amplify modest shocks into severe downturns and leave the economy slow to recover. This paper develops an intermediary-based macro-finance model in which the intermediaries' continuation values depend on the sector's intermediation capacity. Exit removes agents who would otherwise earn high risk premia and erodes a slow-moving stock of intermediation capacity. Lower capacity reduces the continuation values and induces further exit, so shocks to capacity set off a self-reinforcing loop. An analytical characterization shows that intermediary net worth can become trapped at low levels when capacity falls below a critical threshold. The calibrated dynamic model matches unconditional asset-pricing, macroeconomic moments, and crises moments. A benchmark model without exit and capacity dynamics exhibits an amplification-persistence trade-off because high risk premia recapitalize intermediaries too quickly.

JEL: C63, C68, E27, G12, G21, G22, G23

Keywords: financial crises; intermediation capacity; expert exit; crisis persistence; risk premia

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1 Introduction

Financial crises have two defining features. First, relatively modest shocks can generate disproportionately large declines in output, investment, and asset prices, together with sharp spikes in risk premia. This is the amplification dimension of crises. Second, the economy recovers only slowly — even after the initial shock has passed, output remains depressed, intermediary balance sheets remain impaired, and credit spreads stay elevated. This is the persistence dimension of crises. A successful theory of financial crises must therefore explain not only why shocks are amplified but also why their effects last. Intermediary-based macro-finance models provide a natural account of amplification because shocks to intermediary net worth tighten constraints, depress asset prices, and raise risk premia. But this same mechanism creates a difficulty for persistence. Once risk premia rise, constrained intermediaries earn high expected returns on their levered positions, which helps them rebuild net worth quickly. Thus, the endogenous risk-premium channel that makes crises severe also tends to make them short-lived.

This paper studies a mechanism that breaks the tight link between amplification and fast recapitalization. Section 2 builds a quantitative continuous-time macro-finance model in which the deterioration of intermediation capacity induces expert exit, which damages capacity further. The economy is populated by households and more productive experts. Experts intermediate capital, issue outside equity, and face leverage and skin-in-the-game constraints. When intermediation becomes sufficiently unprofitable, experts may choose to leave the sector and become households. Exit is irreversible and removes part of the sector’s active intermediation capacity. Intermediation capacity is an endogenous slow-moving latent state that measures the strength of the active intermediary sector. It aggregates the monitoring expertise, specialized labor, client relationships, etc. that lower the cost of expert intermediation. Capacity rebuilds gradually over time and deteriorates when experts exit. As capacity falls, intermediation becomes less profitable for the experts who remain, their continuation values decline, and future exit incentives rise. Individual exit decisions, therefore, accumulate into persistent aggregate damage.

The central insight is that by the time risk premia are elevated, the economy may already have lost some of the agents able to earn them and repair their balance sheet. In the benchmark model without exit and capacity dynamics, elevated crisis premia speed up recovery because constrained experts rebuild net worth. In the full model, recapitalization also depends on a slow-moving capacity state that exit erodes. As experts leave, active intermediation capacity falls; as capacity falls, staying in the sector becomes less valuable; and as continuation values decline, exit incentives rise further. Because capacity rebuilds slowly, a sector that enters a downturn with impaired capacity remains impaired long after net worth has recovered, so high risk premia coexist with persistent damage to intermediation capacity.

In Section 3, I demonstrate the mechanism in a tractable limiting case. I fix capacity, hold expert risk exposure constant, and study the limit in which experts have a vanishing wealth share. Conditional on remaining in the sector, experts take their risk exposure as given but continue to choose whether to exit when an opportunity arrives. This exercise isolates the race between recapitalization and exit. Allowing exit to feed back into capacity can then generate two self-reinforcing regimes: a low-capacity state with persistent distress and a high-capacity state with recovery. The second contribution is quantitative. In Section 4, I solve the full dynamic model globally and calibrate it to match asset-pricing, macroeconomic, intermediary, and crisis moments. The calibrated model jointly produces high risk premia, realistic crisis amplification, endogenous exit, and slow but eventual recovery of intermediation capacity. It generates an unconditional risk premium of 5.5% and a risk-free rate of 1.5%, close to the empirical values, together with trend output growth of 1.9% and the investment–capital ratio of 7%. Annualized output growth averages -4.0% in crisis states. The resulting damage to intermediation capacity decays with a half-life of about 30 months, comparable to the persistence of the empirical capacity index and to the recovery of credit spreads after financial crises. Leverage is countercyclical, as in the data. I then compare the full model with a benchmark economy with no exit and constant capacity. This model reduces to [Brunnermeier and Sannikov \(2014\)](#) with richer preferences. The benchmark generates substantial amplification where negative shocks reduce expert net worth, tighten constraints, depress asset prices, and raise risk premia. However, the same rise in risk premia that makes crises severe also raises the expected returns of surviving levered experts, allowing them to rebuild net worth quickly. Parameters that increase crisis risk premia and deepen output declines therefore shorten crises, while parameters that raise unconditional risk premia reduce the frequency of crisis states. The comparison shows why the exit–capacity channel matters: by allowing expert exit to erode intermediation capacity, the full model makes high risk premia coexist with persistent impairment of intermediation capacity.

Related Literature This paper is closely related to the intermediary asset-pricing and macro-finance literature that emphasizes the role of financial intermediaries as marginal investors in asset markets. On the theoretical side, [Brunnermeier and Sannikov \(2014\)](#) provides a continuous-time macro-finance model in which intermediary balance-sheet constraints generate endogenous risk, capital misallocation, and crisis dynamics. Related intermediary-friction frameworks include [Di Tella \(2017\)](#), [He and Krishnamurthy \(2013\)](#), [Mendo \(2020\)](#), [Adrian and Boyarchenko \(2012\)](#), all of which stress how leverage or collateral constraints distort the allocation of capital and amplify shocks. My paper builds on this tradition but differs by emphasizing the endogenous exit of experts and the destruction of intermediation capacity as distinct crisis-amplification channels. In my model, intermediary distress does not only tighten balance-sheet constraints contemporaneously; it

can also reduce the future capacity of the financial sector to intermediate risk and capital.

Second, the paper contributes to the quantitative literature that asks whether general-equilibrium macro-finance models with financial frictions can jointly account for macroeconomic fluctuations and asset-pricing moments. The empirical evidence for intermediary-based asset pricing highlights the role of banks, broker-dealers, and hedge funds in pricing assets; see, for example, [He, Kelly, and Manela \(2017\)](#) and [Adrian, Etula, and Muir \(2014\)](#). However, the set of structural models that quantitatively confront both macro and asset-pricing data remains limited. Important contributions include [He and Krishnamurthy \(2019\)](#), [Krishnamurthy and Li \(2025\)](#), [Maxted \(2024\)](#), and related quantitative work by [Gertler, Kiyotaki, and Prestipino \(2020\)](#) and [Elenev, Landvoigt, and Van Nieuwerburgh \(2021\)](#). Relative to [He and Krishnamurthy \(2019\)](#), where bankers solve a myopic portfolio problem, the experts in my model solve an infinite-horizon consumption-portfolio problem. Relative to [Krishnamurthy and Li \(2025\)](#), which emphasizes beliefs and pre-crisis froth, my focus is on the dynamic interaction between expert net worth, exit, and intermediation capacity. The paper is also complementary to [Muir \(2017\)](#), who shows empirically that financial crises are crucial for understanding international variation in risk premia and that intermediary-based asset-pricing models outperform standard representative-agent models with long-run risk, habit, or rare-disaster features.

Third, the paper is related to adjacent literatures on market participation, costly entry, and endogenous exit. [Khorrami \(2016\)](#) studies a limited-participation model with costly entry into the stock market and finds that the implied entry cost can be extremely large relative to wealth. My mechanism is different: rather than focusing on entry into the stock market per se, I study capital misallocation and endogenous exit of experts from the intermediary sector. In this respect the paper is closer to [Eisfeldt, Lustig, and Zhang \(2023\)](#), who introduce endogenous entry and exit in complex asset markets, and to broader literatures on sectoral reallocation and the non-portability of organizational capital. It is also related to [Bigio and D'Avernas \(2021\)](#), who develops a model of state-dependent risk capacity and slow post-crisis recovery. My contribution is to embed exit directly into a global continuous-time macro-finance environment in which eroded intermediation capacity raises the incentive to exit, exit lowers capacity further, and the resulting decline in capacity feeds back into future risk-bearing and asset prices.

Fourth, the paper is connected to the literature on bank runs, banking fragility, and financial regulation, though the mechanism is different. Starting from [Diamond and Dybvig \(1983\)](#), that literature studies fragility arising from liquidity transformation and the role of policy tools such as deposit insurance, liquidity regulation, and capital requirements in preventing destabilizing withdrawals. The modern macro-banking literature, such as [Gertler and Kiyotaki \(2010\)](#) and [Gertler et al. \(2020\)](#), places financial intermediation, banking distress, and policy intervention at the center of aggregate fluctuations. Recent work also studies banking fragility, runs, and the

design of regulation in environments with financial amplification ([Amador and Bianchi \(2024\)](#) and [Amador and Bianchi \(2025\)](#)). My paper differs from this literature because crises are not generated by depositor coordination failures or rollover runs. Instead, the central mechanism is endogenous strategic exit, the resulting destruction of intermediation capacity, and the heightened fragility that arises when net-worth shocks hit a sector whose intermediation capacity is already impaired. The connection to the bank-run and regulation literature is therefore through financial fragility rather than through the run mechanism itself. The analysis in the limiting case of the dynamic model is similar to [Gopalakrishna, Lee, and Papamichalis \(2025\)](#), which shows that belief distortions can push productive agents into permanent crises. In this paper, the limiting exercise isolates locally trapping dynamics generated by exit–capacity interaction, while the quantitative economy uses the same feedback to produce persistent but ultimately transitory impairment of risk-bearing capacity.

2 Model

In this section, I outline the economic model. I study a dynamic, heterogeneous agent production economy with two types of agents: households and experts.

2.1 Environment

Time is continuous and is denoted by $t \in [0, \infty)$. The economy consists of a continuum of two types of price-taking agents: households (h) and experts (e). There is a perishable consumption good and physical capital that agents can invest in. The agents can trade physical capital and a risk-free bond. Production follows AK technology. The net output from each agent type is

$$y_{j,t} = a_j k_{j,t} \quad j \in \{e, h\} \quad (1)$$

where a_j denotes the productivity rate per unit of capital operated by type j . Following [Brunnermeier and Sannikov \(2014\)](#), I assume that $a_e > a_h$. The capital $k_{j,t}$ evolves as

$$\frac{dk_{j,t}}{k_{j,t}} = (\Phi(\iota_{j,t}) - \delta)dt + \sigma dZ_t^k \quad (2)$$

where $\iota_{j,t}$ is the investment rate and Z_t^k is a standard Brownian motion. The parameter σ denotes the exogenous volatility of the capital process. The investment function $\Phi(\cdot)$ is concave and captures the decreasing returns to scale, and δ is the depreciation rate of capital. The depreciation rate is the same for both households and experts. I assume that the investment cost function takes the logarithmic form $\Phi(\iota) = \frac{\log(\phi\iota+1)}{\phi}$ where ϕ is the adjustment cost parameter that controls the

elasticity of the investment technology.¹ Both households and experts can hold capital, and experts can issue outside equity to the households. Households face a no-shorting constraint, and cannot issue outside equity. Experts face a private burden of operating as intermediaries ω_t that depends on the balance sheet capacity governed by

$$dm_t = \mu_{m,t}dt + \sigma_{m,t}dZ_t^m \quad (3)$$

where the drift and diffusion $\mu_{m,t}, \sigma_{m,t}$ are endogenous objects to be explained later, and Z_t^m is a standard Brownian motion that has a correlation of φ with Z_t^k . The capital can be traded at a price q_t that is governed by the differential equation

$$dq_t = q_t \left(\mu_t^q dt + \sigma_t^{q,k} dZ_t^k + \sigma_t^{q,m} dZ_t^m \right) \quad (4)$$

where the functions μ_t^q and $\sigma_t^{q,k}, \sigma_t^{q,m}$ will be determined in equilibrium. As a consequence of the type-specific productivity rate a_j , the return process for the agents is type-dependent. It is given by

$$dR_{j,t} = \underbrace{\left(\frac{a_j - \iota_{j,t}}{q_t} + \mu_t^q + \Phi(\iota_t) - \delta + \sigma \sigma_t^{q,k} + \varphi \sigma \sigma_t^{q,m} \right)}_{\mu_{R,j,t}} dt + \underbrace{\left(\sigma + \sigma_t^{q,k} \right)}_{\sigma_{R,t}^k} dZ_t^k + \underbrace{\sigma_t^{q,m}}_{\sigma_{R,t}^m} dZ_t^m \quad (5)$$

where $\frac{a_j - \iota_{j,t}}{q_t}$ is the dividend yield. Since the market is incomplete due to the presence of financial frictions, each agent type has its own market price of risk $\varsigma_{j,t}$. The wealth of experts is denoted by $w_{e,t}$. Agents have Duffie–Epstein stochastic differential utility with risk aversion $\gamma > 1$ and unit elasticity of intertemporal substitution. The aggregator is

$$f(c, U) = \rho(1 - \gamma)U \left[\log c - \frac{1}{1 - \gamma} \log((1 - \gamma)U) \right]. \quad (6)$$

Let τ_e denote the expert's endogenous exit time, restricted to the arrival times of the Poisson opportunity process. For every bounded stopping time $T \geq t$, the expert's recursive problem can be represented as

$$U_{e,t} = \sup_{\{c_{e,t}, k_{e,t}, \chi_t, \iota_{e,t}, \tau_{e,t}\}} \mathbb{E}_t \left[\int_t^{T \wedge \tau_e} f(c_{e,s}, U_{e,s}) ds + \mathbf{1}_{\{\tau_e > T\}} U_{e,T} + \mathbf{1}_{\{\tau_e \leq T\}} U_{h,\tau_e}^{\text{post}} \right] \quad (7)$$

¹This is a valid investment cost function since $\Phi(0) = 0$, $\Phi' > 0$, and $\Phi'' \leq 0$.

where $U_{h,t}^{post}$ denotes the household utility after exit, defined below. The wealth constraint is

$$\frac{dw_{e,t}}{w_{e,t}} = \underbrace{\mu_{e,t}^w dt + \chi_t \frac{q_t k_{e,t}}{w_{e,t}} (\sigma + \sigma_t^{q,k}) dZ_t^k}_{\sigma_{e,t}^{w,k}} + \underbrace{\chi_t \frac{q_t k_{e,t}}{w_{e,t}} \sigma_t^{q,m} dZ_t^m}_{\sigma_{e,t}^{w,m}} \quad (8)$$

$$\mu_{e,t}^w = r_t - \frac{c_{e,t}}{w_{e,t}} + \frac{q_t k_{e,t}}{w_{e,t}} (\tilde{\mu}_{R,e,t} - r_t) - (1 - \chi_t) \frac{q_t k_{e,t}}{w_{e,t}} \epsilon_{h,t} \quad (9)$$

where $\tilde{\mu}_{R,e,t}$ is the expected return on capital net of intermediation costs. That is, $\tilde{\mu}_{R,e,t} := \mu_{R,e,t} - \omega(m_t)$. They face an outside equity issuance constraint $\chi_t \geq \underline{\chi}$, where χ_t is the fraction of inside equity retained, and a state-dependent capital constraint $\chi_t q_t k_{e,t} \leq w_{e,t} \bar{\sigma} / \|\sigma_{R,t}\|$. The household problem is simpler, as households do not exit, and is given in Appendix A. Throughout, lowercase letters denote individual quantities and uppercase letters denote sectoral aggregates. Let $\mathcal{E}_t$ and $\mathcal{H}_t$ denote the sets of experts and households. Define $W_{e,t} := \int_{i \in \mathcal{E}_t} w_{e,t}(i) di$ and $W_{h,t} := \int_{i \in \mathcal{H}_t} w_{h,t}(i) di$, with analogous definitions for aggregate consumption $C_{e,t}$ and $C_{h,t}$. Expert- and household-operated physical capital are $K_{e,t} := \int_{i \in \mathcal{E}_t} k_{e,t}(i) di$ and $K_{h,t}^{\text{op}} := \int_{i \in \mathcal{H}_t} k_{h,t}^{\text{op}}(i) di$, while aggregate household outside-equity holdings are $E_{h,t}^{\text{out}} := \int_{i \in \mathcal{H}_t} e_{h,t}^{\text{out}}(i) di$. I suppress the individual index i when no confusion can arise. The expert wealth share is

$$\eta_t := \frac{W_{e,t}}{W_{e,t} + W_{h,t}} = \frac{W_{e,t}}{q_t K_t}.$$

The wealth share will serve as an important variable in the model. Experts have an opportunity to exit the market and become households at a Poisson rate τ . The exit decision is irreversible.² When the opportunity arises, an expert draws a switching cost $\lambda \sim U[0, \bar{\lambda}]$ that is i.i.d. across experts and across opportunities. The switching cost λ is nonpecuniary and is expressed in certainty-equivalent wealth units. Upon exit, the expert's wealth is reclassified from the expert sector to the household sector. The continuation value of agent type $j \in \{e, h\}$ is

$$U_{h,t} = \frac{(w_{h,t} \mathcal{J}_{h,t})^{1-\gamma}}{1-\gamma}; \quad U_{e,t} = \frac{(w_{e,t} \mathcal{J}_{e,t})^{1-\gamma}}{1-\gamma}; \quad U_{h,t}^{post}(\lambda) = \frac{(w_{e,t}(1-\lambda) \mathcal{J}_{h,t})^{1-\gamma}}{1-\gamma} \quad (10)$$

where $\mathcal{J}_{j,t}$ is the stochastic investment opportunity process and follows

$$\frac{d\mathcal{J}_{j,t}}{\mathcal{J}_{j,t}} = \mu_{j,t}^{\mathcal{J}} dt + \sigma_{j,t}^{\mathcal{J},k} dZ_t^k + \sigma_{j,t}^{\mathcal{J},m} dZ_t^m$$

²Irreversible individual exit does not imply a secular decline in aggregate expert wealth. With homothetic preferences and wealth-proportional portfolio constraints, the mass of experts is not a separate aggregate state. Exit drains the expert wealth share, while the excess return earned by surviving experts raises it; an invariant distribution of (η, m) is therefore compatible with irreversible exit when the wealth-share drift is inward pointing.

The functions $\mu_{j,t}^{\mathcal{J}}$ and $\sigma_{j,t}^{\mathcal{J}}$ are to be determined in equilibrium.³ Experts choose to exit if the following condition is satisfied

$$\mathcal{J}_{h,t}(1 - \lambda) - \mathcal{J}_{e,t} > 0 \quad (11)$$

Hence, with $z_t := 1 - \frac{\mathcal{J}_{e,t}}{\mathcal{J}_{h,t}}$, the conditional-on-opportunity exit probability is

$$G(z) = \begin{cases} 0, & z \leq 0 \\ z/\bar{\lambda}, & 0 < z \leq \bar{\lambda} \\ 1 & z > \bar{\lambda} \end{cases} \quad (12)$$

I solve for a recursive Markov equilibrium in two state variables, $(\eta_t, m_t) \in \Omega := (0, 1) \times [\underline{m}, \bar{m}]$. Details are relegated to Appendix A.

Asset pricing conditions: The agents choose the investment rate by solving

$$\max_{\iota_{j,t}} \left(\frac{a_j - \iota_{j,t}}{q_t} + \Phi(\iota_{j,t}) \right)$$

Thus, all agents choose the same investment rate $\iota_t = \frac{q_t - 1}{\phi}$. For each type $j \in \{e, h\}$, define the required risk compensation per unit of capital exposure by

$$\epsilon_{j,t} := \langle \mathcal{S}_{j,t}, \sigma_{R,t} \rangle, \quad \sigma_{R,t} = \left(\sigma + \sigma_t^{q,k}, \sigma_t^{q,m} \right).$$

By the martingale condition, the return on expert-operated capital satisfies

$$\begin{aligned} \frac{a_e - \iota_t}{q_t} - \omega(m_t) + \Phi(\iota_t) - \delta + \mu_{q,t} + \sigma \sigma_t^{q,k} + \varphi \sigma \sigma_t^{q,m} - r_t \\ = \chi_t \epsilon_{e,t} + (1 - \chi_t) \epsilon_{h,t} + \Lambda_t^V, \end{aligned} \quad (13a)$$

$$0 \leq \bar{\sigma} - \frac{\chi_t K_t}{\eta_t} \|\sigma_{R,t}\| \quad \perp \quad \Lambda_t^V \geq 0. \quad (13b)$$

The variable Λ_t^V is the shadow value of relaxing the value-at-risk (VaR) constraint, measured in return units per unit of expert-operated capital. When the VaR constraint is slack, $\Lambda_t^V = 0$, and the expected excess return on expert-operated capital equals the equity-share-weighted average of the excess returns required by experts and households, with weights χ_t and $1 - \chi_t$, respectively. When the constraint binds, $\Lambda_t^V \geq 0$ measures the marginal value of relaxing the limit on expert risk exposure. Capacity affects experts through a private intermediation wedge $\omega(m_t)$ where $\omega'(m_t) < 0$. While

³Throughout the main text, $\mathcal{J}_j$ denotes the wealth-scaled multiplier in an individual agent's value function. Appendix A introduces sector capital scaled functions $J_e = \eta q \mathcal{J}_e$ and $J_h = (1 - \eta) q \mathcal{J}_h$ for numerical convenience.

experts are more productive than households, their effective return from operating capital depends on the sector's intermediation capacity. Low capacity increases the wedge and lowers the effective return experts earn on capital holdings. The capacity is endogenous and evolves according to

$$dm_t = (\kappa_m^1 + \kappa_m^2 \kappa_t)(\bar{m} - m_t)dt - \xi_m x(\eta_t, m_t)(m_t - \underline{m})dt + \sigma_m(\eta_t, m_t)dZ_t^m, \quad (14)$$

where κ_t is the expert-operated capital share and

$$x(\eta_t, m_t) := \tau G(z(\eta_t, m_t))$$

is the realized exit hazard. The first term in equation (14) is capacity rebuilding, the second is erosion due to realized exit, and the third is capacity risk. The Brownian shock Z_t^m is correlated with the capital shock Z_t^k with correlation φ . I assume a convenient functional form for the diffusion function $\sigma_m(\eta, m) = \nu \kappa^\beta (\bar{m} - m)(m - \underline{m})$ with $\beta > 1$. Appendix C provides a microfoundation for equation (14), interpreting capacity as intermediary relationship capital that is rebuilt gradually, eroded by expert exit, and exposed to aggregate shocks. Experts exit when the continuation value of becoming a household exceeds the continuation value of remaining an expert after accounting for the idiosyncratic switching cost. The parameter ξ_m measures capacity loss per unit of realized expert exit. The state m_t therefore captures the sector's active intermediation capacity, which is depleted by expert exit and rebuilt only gradually. This slow-moving exit–capacity feedback is the central mechanism: capacity losses persist beyond the shocks that trigger them and shape the sector's response to subsequent shocks. Realized exit erodes m_t ; a lower m_t raises the wedge $\omega(m_t)$, which lowers the value of remaining an expert and raises the exit incentives of those who stay, so the erosion is self-reinforcing and capacity rebuilds slowly. When a negative shock then reduces expert net worth, the state of capacity determines what follows: if capacity is intact, experts benefit from elevated risk premia and rebuild quickly; if capacity has been eroded, the wedge is high, the experts' return advantage is smaller, and the recovery of the sector is governed by the slow rebuilding of m_t rather than by the fast recapitalization of net worth.

The pricing condition of households is given by

$$\frac{a_h - \iota_t}{q_t} + \Phi(\iota_t) - \delta + \mu_{q,t} + \sigma \sigma_t^{q,k} + \varphi \sigma \sigma_t^{q,m} - r_t \leq \epsilon_{h,t} \quad (15)$$

with equality when $\kappa_t < 1$. Define the household capital-pricing slack by

$$s_{h,t} := \epsilon_{h,t} - (\mu_{R,h,t} - r_t) \geq 0.$$

Because households operate the fraction $1 - \kappa_t$ of physical capital, their no-short constraint implies

$$0 \leq 1 - \kappa_t \quad \perp \quad s_{h,t} \geq 0.$$

Subtracting the household return from the return on expert-operated capital gives

$$\frac{a_e - a_h}{q_t} - \omega(m_t) = \chi_t (\epsilon_{e,t} - \epsilon_{h,t}) + \Lambda_t^V + s_{h,t}, \quad (16a)$$

$$0 \leq 1 - \kappa_t \quad \perp \quad \frac{a_e - a_h}{q_t} - \omega(m_t) - \chi_t (\epsilon_{e,t} - \epsilon_{h,t}) - \Lambda_t^V \geq 0. \quad (16b)$$

I restrict attention to equilibria in which $\epsilon_{e,t} \geq \epsilon_{h,t}$ in VaR-slack states, as verified numerically. When the VaR constraint is slack, the outside-equity choice satisfies

$$0 = \min \left\{ \chi_t - \underline{\chi}, \epsilon_{e,t} - \epsilon_{h,t} \right\}. \quad (17)$$

When the VaR constraint binds, the shadow value of relaxing retained expert exposure also enters the equity-retention decision. I focus on the equilibrium regime in which the inside-equity floor binds in VaR-constrained states (i.e., $\chi_t = \underline{\chi}$). When experts operate all capital and the VaR constraint is slack, the inside equity share may have an interior solution, in which case $\epsilon_{e,t} = \epsilon_{h,t}$.

I solve for a decentralized equilibrium in two state variables: the expert wealth share η_t and intermediation capacity m_t . The wealth share dynamics is given by

$$\begin{aligned} \frac{d\eta_t}{\eta_t} &= \mu_t^\eta dt + \sigma_t^{\eta,k} dZ_t^k + \sigma_t^{\eta,m} dZ_t^m \\ \mu_t^\eta &= \frac{(1 - \eta_t)\kappa_t}{\eta_t} \left[(\tilde{\mu}_{R,e,t} - r_t) - (1 - \chi_t)\epsilon_{h,t} \right] - (1 - \chi_t\kappa_t)\epsilon_{h,t} - \omega(m_t)\kappa_t \\ &\quad - \left(\frac{\chi_t\kappa_t}{\eta_t} - 1 \right) \|\sigma_{R,t}\|^2 - \tau G(z_t) \end{aligned} \quad (18)$$

$$\sigma_t^{\eta,k} = \left(\frac{\chi_t\kappa_t}{\eta_t} - 1 \right) (\sigma + \sigma_t^{q,k}) \quad (19)$$

$$\sigma_t^{\eta,m} = \left(\frac{\chi_t\kappa_t}{\eta_t} - 1 \right) \sigma_t^{q,m} \quad (20)$$

Proof: See Appendix B.1.

The last term in equation (18) is the realized exit hazard. The return on expert-operated capital is

$$\tilde{\mu}_{R,e,t} - r_t = \chi_t \epsilon_{e,t} + (1 - \chi_t) \epsilon_{h,t} + \Lambda_t^V.$$

When the VaR constraint is slack, $\Lambda_t^V = 0$. When households operate a positive amount of physical capital, $\kappa_t < 1$, household capital pricing is at equality and $s_{h,t} = 0$. Combining the asset pricing condition then gives

$$\tilde{\mu}_{R,e,t} - r_t = \epsilon_{h,t} + \frac{a_e - a_h}{q_t} - \omega(m_t),$$

regardless of whether the VaR constraint is slack or binding. Two opposing forces shape exit following negative shocks. First, because experts are levered, adverse capital shocks reduce their wealth share. Holding intermediation capacity fixed, the resulting scarcity of expert net worth raises the return advantage and continuation value of surviving experts relative to households, tending to reduce exit. Second, when shocks also impair intermediation capacity—or when capacity has already been eroded by past exit—the intermediation wedge rises, lowering the value of remaining an expert and increasing exit. The recovery of expert net worth depends on the balance between the elevated premium that supports recapitalization and the exit–capacity feedback that impedes it. When capacity is severely impaired, high risk premia continue to support recapitalization but may be insufficient to offset the intermediation wedge and exit.

3 Net worth trap

This section studies a limiting case of the model in Section 2 that isolates the exit–capacity mechanism. I fix a candidate capacity level $m_0 \in [\underline{m}, \bar{m}]$, hold capacity at m_0 , and study the boundary $\eta \downarrow 0$ under a fixed risk exposure policy. Conditional on remaining in the sector, experts take the risk exposure as given but endogenously choose whether to exit. The exercise therefore holds local amplification fixed and asks whether recapitalization or exit dominates as capacity varies.

I begin by deriving from the wealth-share dynamics in (18) a local identity that isolates the recapitalization channel. Consider a crisis state in which households operate a positive amount of capital, $\kappa_t < 1$.⁴ The household capital-pricing condition then holds with equality, so $\epsilon_{h,t} = \mu_{R,h,t} - r_t$ is the equilibrium expected excess return on household-operated capital rather than a shadow required return. I use $\epsilon_{h,t}$ as a local measure of crisis amplification.

Proposition 1 (Amplification and local recapitalization). *Fix a crisis state with $\kappa_t < 1$ in the economy without exit or a capacity wedge, $\tau = 0$ and $\omega \equiv 0$. Define*

$$\Delta a := a_e - a_h, \quad \ell_{e,t} := \frac{\chi_t \kappa_t}{\eta_t} > 1, \quad v_t^2 := \|\sigma_{R,t}\|^2.$$

Consider a local comparative static that varies the household risk premium $\epsilon_{h,t}$ while holding the

⁴This condition holds throughout the crisis region of the calibrated model in Section 4.

local allocation $(\eta_t, \kappa_t, \ell_{e,t})$, the capital price q_t , and return volatility v_t fixed. The local drift of the log expert wealth share is

$$\mu_t^B = \frac{(1 - \eta_t)\kappa_t}{\eta_t} \frac{\Delta a}{q_t} + (\ell_{e,t} - 1)\epsilon_{h,t} - \frac{1}{2}(\ell_{e,t} - 1)(\ell_{e,t} + 1)v_t^2.$$

It follows that

$$\frac{\partial \mu_t^B}{\partial \epsilon_{h,t}} = \ell_{e,t} - 1 > 0.$$

If the leverage constraint $\ell_{e,t}v_t \leq \bar{\sigma}$ is imposed, then

$$\mu_t^B \geq \underline{\mu}_t^B(\epsilon_{h,t}) := \frac{(1 - \eta_t)\kappa_t}{\eta_t} \frac{\Delta a}{q_t} + (\ell_{e,t} - 1)\epsilon_{h,t} - \frac{\bar{\sigma}^2}{2} \left(1 - \frac{1}{\ell_{e,t}^2}\right).$$

The bound is attained when the leverage constraint binds and is strictly increasing in $\epsilon_{h,t}$.

Proposition 1 shows that, holding the local capital allocation, exposure, and return volatility fixed, a higher crisis risk premium raises the recapitalization drift of experts at rate $(\ell_{e,t} - 1)$, their excess exposure. The following corollary translates this effect into expected recovery time under the constant-coefficient approximation.

Corollary 1 (Local amplification–persistence trade-off). *Fix any wealth-share level $\bar{\eta} > \eta_t$ within the local neighborhood of the crisis state, and define its first-passage time by*

$$T_{\bar{\eta}} := \inf \{s \geq 0 : \eta_{t+s} \geq \bar{\eta}\}.$$

Under the approximations in Proposition 1, the log expert wealth share evolves as

$$d \log \eta_{t+s} = \mu_t^B ds + (\ell_{e,t} - 1)v_t dW_s,$$

where W_s is a standard Brownian motion. If $\mu_t^B > 0$, then

$$\mathbb{E}_t[T_{\bar{\eta}}] = \frac{\log(\bar{\eta}/\eta_t)}{\mu_t^B}.$$

Moreover,

$$\frac{\partial \mathbb{E}_t[T_{\bar{\eta}}]}{\partial \epsilon_{h,t}} = -\frac{\log(\bar{\eta}/\eta_t)(\ell_{e,t} - 1)}{(\mu_t^B)^2} < 0.$$

Proof of Proposition 1 and Corollary 1: See Appendix B.8.

Intuitively, experts hold exposure $\ell_{e,t}$ per dollar of net worth and earn the crisis premium on their excess exposure $\ell_{e,t} - 1$. The leverage constraint caps the volatility of expert net worth, so a higher

crisis premium raises the lower bound on the recapitalization drift at rate $\ell_{e,t} - 1$ for any return volatility consistent with the cap. When the constraint binds, the bound is attained and a larger crisis premium raises the conditional log drift and lowers expected recovery time. While this analysis holds equilibrium prices and portfolios locally fixed, in the full model, we get

$$\mu_t^F = \mu_t^B - \frac{\kappa_t}{\eta_t} \omega(m_t) - \tau G(z_t),$$

Exit and impaired capacity directly reduce recapitalization of the expert sector. Although equilibrium risk premia may rise in response, they need not offset these drags, so a high crisis premium need not imply rapid recovery. The remainder of this section develops the boundary analysis. Proposition 2 establishes a conditional capacity threshold, Corollary 2 characterizes in closed form how the exit–capacity channel shifts the boundary recovery frontier, Proposition 3 extends the result to general switching-cost distributions and intermediation wedges, and Proposition 4 characterizes the locally trapping and locally escaping regimes of the reduced boundary system.

For a fixed capacity level, the local expert wealth-share dynamics take the form

$$\frac{d\eta_t}{\eta_t} = \mu_0(m_0) dt + \sigma_0(m_0) dZ_t.$$

Applying Itô's lemma gives

$$d \log \eta_t = \left[\mu_0(m_0) - \frac{1}{2} \sigma_0(m_0)^2 \right] dt + \sigma_0(m_0) dZ_t.$$

The sign of the log drift governs the local behavior of expert net worth near the boundary. To express this condition compactly, define the boundary recovery index

$$D_0(m_0) := \frac{2\mu_0(m_0)}{\sigma_0(m_0)^2}. \quad (21)$$

The log drift can then be written as

$$\mu_0(m_0) - \frac{1}{2} \sigma_0(m_0)^2 = \frac{1}{2} \sigma_0(m_0)^2 [D_0(m_0) - 1].$$

Thus, $D_0(m_0) > 1$ corresponds to positive local log growth, $D_0(m_0) < 1$ to negative local log growth, and $D_0(m_0) = 1$ to the critical case.

Definition 1 (Boundary classification). *For the local wealth-share dynamics at capacity m_0 , the expert net-worth boundary is locally escaping if $D_0(m_0) > 1$, critical if $D_0(m_0) = 1$, and locally*

trapping if $D_0(m_0) < 1$. This classification describes recovery to a fixed interior wealth share as the initial expert wealth share approaches zero.⁵

Fix any $0 < \eta_0 < \bar{\eta}$, and let

$$T_{\bar{\eta}} := \inf\{t \geq 0 : \eta_t \geq \bar{\eta}\}$$

denote the first time the expert wealth share reaches $\bar{\eta}$. Under the local wealth-share dynamics at capacity m_0 ,

$$\Pr_{\eta_0}(T_{\bar{\eta}} < \infty) = \begin{cases} 1, & D_0(m_0) \geq 1, \\ \left(\frac{\eta_0}{\bar{\eta}}\right)^{1-D_0(m_0)}, & D_0(m_0) < 1. \end{cases}$$

When the boundary is locally trapping, the probability of recovery converges to zero as $\eta_0 \downarrow 0$. At the critical value $D_0(m_0) = 1$, recovery occurs almost surely, but the expected recovery time is infinite. These results are established in Step 4 of the proof of Proposition 2.

Assumption 1 (Boundary conditions). *For each $m \in (\underline{m}, \bar{m})$, the wealth-scaled value multipliers $\mathcal{J}_e$, $\mathcal{J}_h$, and the capital price q are C^2 near $(0, m)$ with bounded first and second derivatives near $(0, m)$, and their boundary limits extend continuously to $m \in [\underline{m}, \bar{m}]$. The feasible set of the expert continuation problem holds boundary exposure fixed as $\eta \downarrow 0$:*

$$\chi \longrightarrow \underline{\chi}, \quad \frac{\chi \kappa}{\eta} \longrightarrow \frac{\bar{\sigma}}{\sigma}, \quad \frac{\kappa}{\eta} \longrightarrow \ell_0 := \frac{\bar{\sigma}}{\underline{\chi} \sigma}.$$

Experts take risk exposure as fixed but retain the option to exit, with $\bar{\sigma}$ determining exposure even where the VaR constraint would be slack in the full equilibrium. Assume $1 + \phi a_h > 0$, $\bar{\sigma} > \sigma$, $\gamma > 1$, and $0 < \bar{\lambda} < 1$, so that

$$q_0 = \frac{1 + \phi a_h}{1 + \phi \rho} > 0.$$

The intermediation wedge is

$$\omega(m) = \bar{\omega} \frac{\bar{m} - m}{\bar{m} - \underline{m}}, \quad m \in [\underline{m}, \bar{m}].$$

The boundary analysis treats m_0 as a parameter.

Lemma 1 (Boundary recapitalization). *Under Assumption 1, with equations (18)–(20) evaluated under the boundary risk exposure, the limiting drift and diffusion of the expert wealth share at*

⁵Note that it does not impose an absorbing boundary at $\eta = 0$ or imply the existence of an invariant distribution.

capacity m_0 are

$$\mu_0(m_0) := \lim_{\eta \downarrow 0} \mu^\eta(\eta, m_0) = \underbrace{\ell_0 \pi(m_0) + (\gamma - 1) \sigma \sigma_0}_{\mathcal{R}(m_0)} - x_0(m_0),$$

and

$$\sigma_0 := \lim_{\eta \downarrow 0} \sigma^\eta(\eta, m_0) = \bar{\sigma} - \sigma,$$

where

$$\ell_0 := \frac{\bar{\sigma}}{\underline{\chi} \sigma}, \quad \pi(m) := \frac{\Delta a}{q_0} - \omega(m), \quad q_0 := \frac{1 + \phi a_h}{1 + \phi \rho},$$

and

$$z_0(m) := 1 - \frac{\mathcal{J}_{e,0}(m)}{\mathcal{J}_{h,0}(m)}, \quad x_0(m) := \tau G(z_0(m))$$

is the realized boundary exit hazard. If the boundary exit gap is small and interior, $0 < z_0(m_0) < \bar{\lambda}$, then

$$x_0(m_0) = \frac{\tau}{\bar{\lambda} \rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m_0) \right] + O(z_0(m_0)^2).$$

A convenient first-order extension across the no-exit margin is

$$x_0^{\text{lin}}(m) := \frac{\tau}{\bar{\lambda} \rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m) \right]^+.$$

Proof: See Appendix B.2.

Lemma 1 decomposes boundary expert-net-worth growth into a recapitalization force and an exit force. The recapitalization force

$$\mathcal{R}(m) = \ell_0 \pi(m) + (\gamma - 1) \sigma \sigma_0$$

reflects experts' levered effective return advantage and their exposure to aggregate risk. The realized exit hazard $x_0(m)$ subtracts directly from expert wealth-share growth. Higher capacity lowers the intermediation wedge, raises $\pi(m)$, and increases the continuation value of remaining an expert; it therefore raises recapitalization directly and reduces exit. The relevant object for boundary escape is the log-growth rate $\mu_0(m) - \frac{1}{2} \sigma_0^2$, rather than $\mu_0(m)$ alone. Thus, volatility determines how large the drift must be for the boundary to be escaping.

Corollary 2 (Boundary recovery frontier). *Under Assumption 1, let $\epsilon_{h,0} := \gamma \sigma^2$ denote the household boundary risk premium.⁶ Absent exit and the capacity wedge, $x_0 \equiv 0$ and $\omega \equiv 0$, the boundary*

⁶The derivation is in Appendix B.2.

recovery index is

$$D_0^B(\epsilon_{h,0}) = \frac{2\ell_0\Delta a}{q_0\sigma_0^2} + \frac{2\epsilon_{h,0}}{\sigma\sigma_0} - \frac{2\sigma}{\sigma_0}.$$

Holding σ and σ_0 fixed,

$$\frac{\partial D_0^B}{\partial \epsilon_{h,0}} = \frac{2}{\sigma\sigma_0} > 0.$$

For any $0 < \eta_0 < \eta_N$, define

$$T_{\eta_N} := \inf\{t \geq 0 : \eta_t \geq \eta_N\}.$$

If $D_0^B > 1$, the expected time required to reach η_N is

$$\mathbb{E}_{\eta_0}[T_{\eta_N}] = \frac{2\log(\eta_N/\eta_0)}{\sigma_0^2(D_0^B - 1)},$$

which is strictly decreasing in $\epsilon_{h,0}$.

With exit and the capacity wedge, the boundary recovery index at a given capacity level and boundary premium becomes

$$D_0^F(m; \epsilon_{h,0}) = D_0^B(\epsilon_{h,0}) - \frac{2}{\sigma_0^2} [\ell_0\omega(m) + x_0(m)].$$

Exit and impaired capacity therefore shift the recovery frontier weakly downward, and strictly so whenever $\ell_0\omega(m) + x_0(m) > 0$. The boundary is no longer locally escaping when

$$\ell_0\omega(m) + x_0(m) \geq \frac{\sigma_0^2}{2} [D_0^B(\epsilon_{h,0}) - 1].$$

Proof: Immediate from Lemma 1, the boundary limits in Appendix B.2 ($\kappa_t/\eta_t \rightarrow \ell_0$, $\chi_t\kappa_t/\eta_t \rightarrow \bar{\sigma}/\sigma$, $v_t^2 \rightarrow \sigma^2$, $\epsilon_{h,0} \rightarrow \gamma\sigma^2$), and the definition (21).

The corollary shows that higher risk premia accelerate expert recapitalization when exit and the capacity wedge are absent. Exit and impaired capacity offset this force and, when sufficiently large, turn a locally recovering boundary into a locally trapping one.

Proposition 2 (Capacity threshold). *Under Assumption 1, let $D_0(m)$ denote the boundary recovery index in (21), evaluated using the exact exit hazard $x_0(m)$. The function $D_0(m)$ is continuous and strictly increasing in capacity. Suppose $D_0(\underline{m}) < 1 < D_0(\bar{m})$. Then there exists a unique $m^\star \in (\underline{m}, \bar{m})$ such that*

$$D_0(m^\star) = 1.$$

Moreover,

$$D_0(m) < 1 \quad \text{for } m < m^*, \quad D_0(m) > 1 \quad \text{for } m > m^*.$$

Thus, the boundary is locally trapping below m^* , critical at m^* , and locally escaping above m^* . In particular, when $m < m^*$, the probability of reaching any fixed interior wealth share converges to zero as the initial expert wealth share approaches zero.

If m^* lies in the small interior-exit region, let $z_\star := z_0(m^*)$. Then

$$m^* = \tilde{m}^* + O(z_\star^2),$$

where

$$\tilde{m}^* = \bar{m} - \frac{\bar{m} - \underline{m}}{\bar{\omega}} \left[\frac{\Delta a}{q_0} - \frac{\frac{1}{2}\sigma_0^2(1+b\gamma) - (\gamma-1)\sigma\sigma_0}{(1+b)\ell_0} \right], \quad b := \frac{\tau}{\bar{\lambda}\rho}. \quad (22)$$

Proof: See Appendix B.3.

Equation (22) is a first-order approximation to the exact threshold characterized by $D_0(m^*) = 1$. Its comparative statics are

$$\begin{aligned} \frac{\partial \tilde{m}^*}{\partial \tau} &= \frac{\bar{m} - \underline{m}}{\bar{\omega}\bar{\lambda}\rho} \frac{(\gamma-1)\sigma_0 \left(\sigma + \frac{1}{2}\sigma_0 \right)}{\ell_0(1+b)^2} > 0, \\ \frac{\partial \tilde{m}^*}{\partial \Delta a} &= -\frac{\bar{m} - \underline{m}}{\bar{\omega}q_0} < 0, \\ \frac{\partial \tilde{m}^*}{\partial \bar{\sigma}} &= \frac{\bar{m} - \underline{m}}{\bar{\omega}} \frac{\chi\sigma}{(1+b)\bar{\sigma}^2} \left[(1+b\gamma)\sigma_0 \left(\sigma + \frac{1}{2}\sigma_0 \right) - (\gamma-1)\sigma^2 \right] \geq 0. \end{aligned}$$

A higher exit-opportunity rate τ raises the capacity threshold by increasing the first-order outflow of experts from the sector. A larger productivity advantage Δa lowers the threshold by strengthening experts' effective return advantage, holding a_h , and hence q_0 , fixed. The effect of $\bar{\sigma}$ is ambiguous: greater expert exposure raises the returns from remaining in the sector but also increases net-worth volatility and, consequently, the drift required for boundary escape.⁷

Corollary 3 (Exact capacity multiplier). *Away from the kinks of the exit rule, capacity raises the exact fixed-capacity boundary drift according to*

$$\frac{\partial \mu_0(m)}{\partial m} = -\ell_0\omega'(m) [1 + \mathcal{M}_0(m)] > 0, \quad (23)$$

⁷These comparative statics pertain to the first-order threshold $\tilde{m}^*$; the exact threshold is defined implicitly by $D_0(m^*) = 1$.

where

$$\mathcal{M}_0(m) := \frac{\tau G'(z_0(m)) e^{\Delta_0(m)}}{\rho - \mathcal{V}_\Delta(\Delta_0(m))} \geq 0, \quad \Delta_0(m) := \log \frac{\mathcal{J}_{e,0}(m)}{\mathcal{J}_{h,0}(m)}, \quad (24)$$

and $\mathcal{V}$ is the exit-option term defined in equation (44). At the kinks, the same monotonicity holds in the one-sided sense. In the no-exit and saturated-exit regions, $G'(z_0) = 0$ away from the kinks, so $\mathcal{M}_0(m) = 0$ and only the direct return-wedge channel remains.

Proof: See Appendix B.4.

Corollary 3 decomposes the effect of capacity on expert recapitalization into two channels. Higher capacity lowers the intermediation wedge and directly raises experts' effective return advantage. It also increases the value of remaining an expert and thereby reduces exit. The total effect on the boundary drift equals the direct effect multiplied by $1 + \mathcal{M}_0(m)$. The option value of future exit opportunities dampens the second channel. Because $\mathcal{V}_\Delta \leq 0$, it enlarges the denominator in (24) and reduces $\mathcal{M}_0(m)$. It therefore affects the magnitude, but not the sign, of the exit response. Setting $\mathcal{V}_\Delta = 0$ removes this feedback and recovers the multiplier used in the first-order approximation.

Robustness of the capacity threshold. Proposition 2 uses two convenient functional forms from Section 2: uniformly distributed switching costs and an intermediation wedge that is affine in capacity. Neither is essential. The next result allows for a general smooth switching-cost distribution and any strictly decreasing wedge. The boundary value gap remains unique, exit declines with capacity, and the recovery index rises with capacity, yielding a unique capacity threshold whenever $D_0(\underline{m}) < 1 < D_0(\bar{m})$. The capacity multiplier in Corollary 3 also extends directly: the uniform density is replaced by the switching-cost density evaluated at the exit margin.

Assumption 2 (General switching costs and capacity wedge). *Switching costs are drawn from a distribution F_λ supported on $[0, \bar{\lambda}]$ with $0 < \bar{\lambda} < 1$, $F_\lambda(0) = 0$, and $F_\lambda(\bar{\lambda}) = 1$, where F_λ is continuously differentiable with density $f_\lambda \geq 0$. The intermediation wedge $\omega : [\underline{m}, \bar{m}] \rightarrow \mathbb{R}_+$ is C^1 and strictly decreasing, $\omega'(m) < 0$. This replaces the uniform switching-cost distribution of Section 2 and the affine wedge of Assumption 1. All remaining conditions of Assumption 1 are maintained.*

Under Assumption 2, the boundary exit-option term generalizes equation (44) to

$$\mathcal{V}_F(\Delta) := \frac{\tau}{1-\gamma} \int_0^{\widehat{z}(\Delta)} \left[\left((1-\lambda)e^{-\Delta} \right)^{1-\gamma} - 1 \right] dF_\lambda(\lambda), \quad \widehat{z}(\Delta) := \min \{ (1-e^\Delta)^+, \bar{\lambda} \}, \quad (25)$$

with derivative $\mathcal{V}_{F,\Delta}$, and reduces to $\mathcal{V}$ when F_λ is uniform.

Proposition 3 (General capacity threshold). *Under Assumption 2, the following statements hold.*

(i) Existence and uniqueness. For each $m \in [\underline{m}, \bar{m}]$, the exact boundary value-gap equation

$$\rho \Delta_0(m) = \ell_0 \pi(m) - \frac{\gamma}{2} \sigma_0^2 + \mathcal{V}_F(\Delta_0(m)) \quad (26)$$

has a unique solution. The associated exit hazard is

$$x_0(m) = \tau F_\lambda(\widehat{z}(\Delta_0(m))).$$

(ii) Monotonicity. The value gap $\Delta_0(m)$ is strictly increasing in capacity, while the exit cutoff $\widehat{z}(\Delta_0(m))$ and exit hazard $x_0(m)$ are non-increasing.

(iii) Capacity multiplier. The boundary drift $\mu_0(m)$ and recovery index $D_0(m)$ are continuous and strictly increasing in capacity. In the interior-exit region, where $0 < z_0(m) < \bar{\lambda}$,

$$\frac{\partial \mu_0(m)}{\partial m} = -\ell_0 \omega'(m) [1 + \mathcal{M}_F(m)], \quad \mathcal{M}_F(m) := \frac{\tau f_\lambda(z_0(m)) e^{\Delta_0(m)}}{\rho - \mathcal{V}_{F,\Delta}(\Delta_0(m))} \geq 0. \quad (27)$$

In the interiors of the no-exit and saturated-exit regions, $x_0(m)$ is locally constant and

$$\frac{\partial \mu_0(m)}{\partial m} = -\ell_0 \omega'(m).$$

Capacity then affects the boundary drift only through the intermediation wedge. At the kinks of the exit rule, the same monotonicity holds in terms of the corresponding one-sided derivatives.

(iv) Capacity threshold. If

$$D_0(\underline{m}) < 1 < D_0(\bar{m}),$$

there exists a unique $m^\star \in (\underline{m}, \bar{m})$ satisfying $D_0(m^\star) = 1$. The boundary is locally trapping for $m < m^\star$, critical at $m = m^\star$, and locally escaping for $m > m^\star$.

Proof: See Appendix B.5.

Proposition 3 shows that the capacity threshold does not depend on the uniform switching-cost distribution or the affine intermediation wedge. With any smooth cost distribution and any strictly decreasing wedge, the boundary value gap remains unique, exit declines with capacity, and the recovery index rises with capacity. Whenever $D_0(\underline{m}) < 1 < D_0(\bar{m})$, these properties yield a unique threshold separating locally trapping from locally escaping boundaries.

Equation (27) shows that the endogenous-exit multiplier retains the same structure. The uniform density $1/\bar{\lambda}$ is replaced by the switching-cost density evaluated at the exit margin. Because

$\rho - \mathcal{V}_{F,\Delta} \geq \rho$, the continuation value of future exit opportunities dampens the exit response to capacity but cannot reverse its sign.

Corollary 4 (First-order threshold with general switching costs). *Under Assumption 2, suppose that f_λ is Lipschitz continuous at zero, and define*

$$b_F := \frac{\tau f_\lambda(0)}{\rho}.$$

Near the no-exit margin, the boundary exit hazard satisfies

$$x_0(m) = b_F \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m) \right]^+ + O(z_0(m)^2).$$

Suppose that

$$D_0(\underline{m}) < 1 < D_0(\bar{m})$$

and that the threshold $m^\star$ from Proposition 3(iv) lies in the small interior-exit region. Define $z_\star := z_0(m^\star)$. If the expression below lies in the range of ω , then

$$m^\star = \tilde{m}_F^\star + O(z_\star^2),$$

where

$$\tilde{m}_F^\star := \omega^{-1} \left(\frac{\Delta a}{q_0} - \frac{\frac{1}{2} \sigma_0^2 (1 + b_F \gamma) - (\gamma - 1) \sigma \sigma_0}{(1 + b_F) \ell_0} \right).$$

When F_λ is uniform on $[0, \bar{\lambda}]$ and ω is affine, $f_\lambda(0) = 1/\bar{\lambda}$, so $b_F = b$ and $\tilde{m}_F^\star$ reduces to the closed-form threshold in (22).

Proof: See Appendix B.6.

Corollary 4 shows that the density of switching costs at zero is a sufficient statistic for exit at first order. It enters the threshold through $b_F = \tau f_\lambda(0)/\rho$. Near the expert-net-worth boundary, experts at the exit margin face nearly zero switching costs. The prevalence of low-cost experts therefore determines how strongly exit slows recapitalization. The remainder of the cost distribution is irrelevant at first order. Distributions with the same $f_\lambda(0)$ yield the same first-order threshold, so the uniform distribution is a normalization rather than a substantive restriction.

The analysis so far treats capacity as given. For each m , Proposition 2 determines the boundary exit schedule $x_0(m)$ and whether experts recapitalize. I now use this schedule to study the feedback from exit to capacity. In particular, I substitute $x_0(m)$ into the reduced capacity law. Because the exit decision is solved separately at each capacity level, agents do not internalize the subsequent

movement of m . The resulting fixed points are steady states of the reduced exit–capacity dynamics. Their purpose is to show that the mechanism can support multiple capacity regimes.

At the boundary, $\kappa = \ell_0\eta + o(\eta)$, so expert activity vanishes as $\eta \downarrow 0$. The intermediation-driven rebuilding and capacity risk in (14) therefore vanish as well. Letting $\kappa_m := \kappa_m^1$, the remaining capacity dynamics are

$$\dot{m} = g(m) := \kappa_m(\bar{m} - m) - \xi_m x_0(m)(m - \underline{m}), \quad (28)$$

where $x_0(m) = \tau G(z_0(m))$ is determined by the exact boundary value-gap equation (47). The steady states of (28) are equivalently the fixed points of

$$F(m) := \frac{\kappa_m \bar{m} + \xi_m x_0(m) \underline{m}}{\kappa_m + \xi_m x_0(m)}. \quad (29)$$

Proposition 4 (Multiple capacity regimes). *Under Assumption 1, let $\kappa_m := \kappa_m^1 > 0$ and $\xi_m > 0$. Define the candidate low-capacity steady state*

$$\widehat{m}_L := \frac{\kappa_m \bar{m} + \xi_m \tau \underline{m}}{\kappa_m + \xi_m \tau}. \quad (30)$$

Suppose that the boundary exit cutoff satisfies

$$z_0(\widehat{m}_L) > \bar{\lambda}, \quad z_0(\bar{m}) < 0. \quad (31)$$

Then $\widehat{m}_L$ and $\bar{m}$ are locally asymptotically stable steady states of the reduced capacity law (28). Exit is saturated at the low-capacity steady state, $x_0(\widehat{m}_L) = \tau$, and absent at the high-capacity steady state, $x_0(\bar{m}) = 0$. In addition, at least one interior steady state exists in $(\widehat{m}_L, \bar{m})$. If g has a unique interior zero, denote it by $\widehat{m}_U$.⁸ The interior steady state $\widehat{m}_U$ is unstable and separates the two capacity basins:

$$m_t \longrightarrow \begin{cases} \widehat{m}_L, & m_0 \in (\widehat{m}_L, \widehat{m}_U), \\ \bar{m}, & m_0 \in (\widehat{m}_U, \bar{m}). \end{cases}$$

Suppose, in addition, that

$$D_0(\widehat{m}_L) < 1 < D_0(\bar{m}). \quad (32)$$

The expert-net-worth boundary is then locally trapping at the low-capacity steady state and locally escaping at the full-capacity steady state. To express this condition in terms of exit, define the

⁸Uniqueness is imposed directly on the endogenous exit schedule rather than derived from restrictions on primitives. It holds in the numerical illustrations in Figure 1. With multiple interior steady states, their stability alternates, while the two outer steady states remain locally stable.

critical exit rate

$$\bar{x}_0(m) := \ell_0 \pi(m) + (\gamma - 1) \sigma \sigma_0 - \frac{1}{2} \sigma_0^2.$$

Because $x_0(\hat{m}_L) = \tau$ and $x_0(\bar{m}) = 0$, condition (32) is equivalent to

$$\tau > \bar{x}_0(\hat{m}_L), \quad \bar{x}_0(\bar{m}) > 0.$$

Proof: See Appendix B.7.

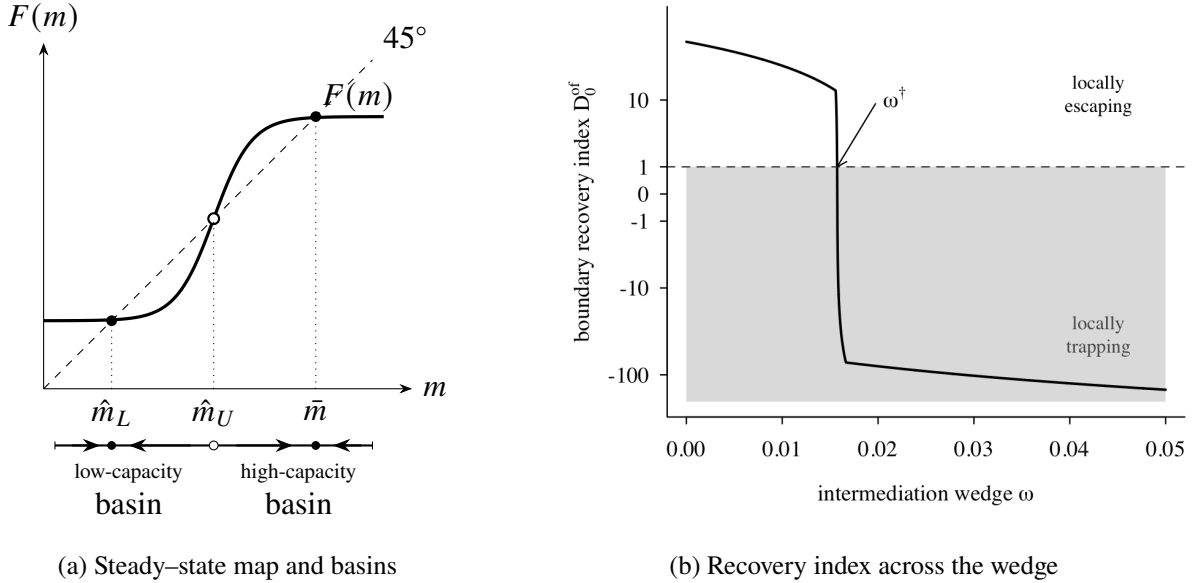


Figure 1. (a) Schematic fixed-point map $F(m)$ for the reduced quasi-static boundary system. Filled dots are locally stable steady states of the *capacity law*: $\hat{m}_L$ has saturated exit and $\bar{m}$ has no exit. Under condition (32), these capacity states are associated with a locally trapping and a locally escaping expert-net-worth boundary, respectively. The open dot is the unique unstable separator $\hat{m}_U$ under the single-crossing condition in Proposition 4. (b) The boundary recovery index under the illustrative option-free schedule (33), plotted against the intermediation wedge. Parameters are $\Delta a = 0.02$, $a_h = 0.06$, $\rho = 0.04$, $\sigma = 0.06$, $\phi = 8$, $\bar{\sigma} = 0.12$, $\chi = 0.5$, $\gamma = 5$, $\tau = 0.15$, and $\bar{\lambda} = 0.10$. Both panels are theoretical illustrations rather than the baseline quantitative calibration.

For the numerical illustrations in Figures 1–2, I use the nonlinear option-free boundary schedule

$$x_0^{\text{of}}(m) := \tau G \left(1 - \exp \left\{ \frac{1}{\rho} \left[\ell_0 \pi(m) - \frac{\gamma}{2} \sigma_0^2 \right] \right\} \right). \quad (33)$$

Unlike the linearized schedule, this specification retains the nonlinear mapping $z = 1 - e^\Delta$ from the value gap to the exit cutoff. It therefore preserves both the no-exit and saturated-exit regions, but omits the exit-option term $\mathcal{V}$. Proposition 4 is stated for the exact schedule $x_0(m)$. It shows how the capacity threshold for expert recapitalization can generate multiple regimes when exit feeds back into intermediation capacity. When capacity is low, the wedge is high and the value of remaining an expert falls. If the cutoff exceeds the switching-cost support, exit saturates at $x_0 = \tau$;

the resulting erosion of capacity validates the low-capacity fixed point $\widehat{m}_L$. At full capacity, experts prefer to remain in the sector, exit shuts down, and mean reversion supports $\bar{m}$. Under a unique interior crossing, $\widehat{m}_U$ separates the low- and high-capacity basins.

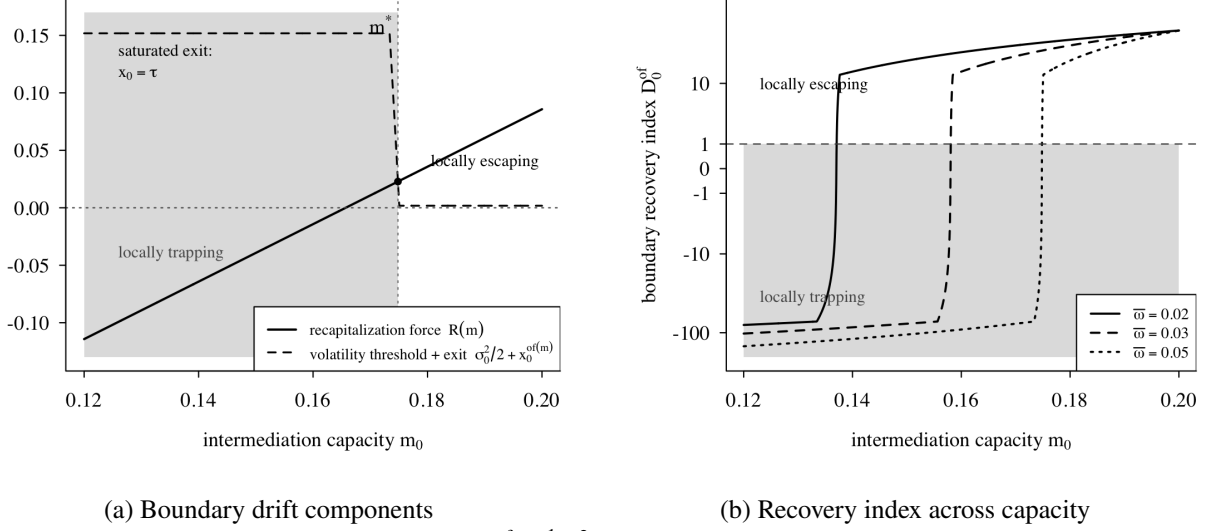


Figure 2. (a) Components of the local log-drift $\mu_0^{\text{of}} - \frac{1}{2}\sigma_0^2$ under the illustrative option-free schedule (33). The solid line is the recapitalization force $R(m) = \ell_0\pi(m) + (\gamma - 1)\sigma\sigma_0$; the dashed line is the sum of the volatility threshold $\sigma_0^2/2$ and the exit hazard $x_0^{\text{of}}(m)$. (b) The corresponding option-free boundary recovery index $D_0^{\text{of}} = 2\mu_0^{\text{of}}/\sigma_0^2$ across capacity for three wedge ceilings $\bar{\omega}$. The shaded region is locally trapping and the unshaded region is locally escaping. Parameters are as in Figure 1, with $\underline{m} = 0.12$ and $\bar{m} = 0.20$. Both panels are theoretical illustrations rather than the baseline calibration.

The distinction between capacity stability and expert-net-worth stability is important. Both $\widehat{m}_L$ and $\bar{m}$ are locally stable fixed points of the one-dimensional capacity law. The expert-net-worth boundary is attracting at $\widehat{m}_L$ only when $D_0(\widehat{m}_L) < 1$, and it is repelling at $\bar{m}$ when $D_0(\bar{m}) > 1$. Panel (b) of Figure 1 plots the illustrative boundary recovery index against the intermediation wedge. A low wedge preserves experts' effective return advantage and supports local escape. As the wedge rises, both the direct recapitalization force and the value of remaining an expert fall. Exit therefore rises and lowers the boundary drift further. Once the recovery index crosses one, the boundary changes from locally escaping to locally trapping.

4 Quantitative analysis

Section 3 provides an analytical characterization of the exit–capacity mechanism near the expert-net-worth boundary. I now return to the full dynamic model introduced in Section 2, in which portfolio choice is endogenous and (η, m) evolve jointly, and solve its global equilibrium quantitatively. The quantitative exercise asks whether the exit–capacity feedback can generate persistent crises followed by slow recovery in an ergodic equilibrium. I compute the global solution

using the neural-network method of [Gopalakrishna and Wu \(2026\)](#). Appendix D describes the numerical procedure.

4.1 Calibration

Table 1 summarizes the calibrated parameters. I follow a hybrid calibration strategy: some parameters are taken from the literature, and the rest are calibrated to match empirical moments. Define the model's crisis region as

$$C := \{(\eta, m) : \chi(\eta, m) = \underline{\chi}, \kappa(\eta, m) < 1\}, \quad C_t := \mathbf{1}\{(\eta_t, m_t) \in C\}.$$

Thus, in a crisis state the skin-in-the-game constraint binds and households operate a positive share of aggregate capital. Write $\bar{a}_t := \kappa_t a_e + (1 - \kappa_t) a_h$ and $Y_t := \bar{a}_t K_t$ for aggregate productivity and output. Following [He and Krishnamurthy \(2019\)](#), Table 2 reports

$$g_Y^C := \mathbb{E}[\log Y_t - \log Y_{t-1} \mid C_t = 1],$$

where observations are annualized and the expectation is taken under the stationary distribution. This statistic summarizes output amplification in crisis states. Let $B_t := q_t K_{e,t}$ denote the market value of capital intermediated by experts. Since $K_{e,t} = \kappa_t K_t$, the model counterpart of intermediated credit relative to output is

$$b_t := \frac{B_t}{Y_t} = \frac{q_t \kappa_t}{\bar{a}_t}.$$

Following [Krishnamurthy and Li \(2025\)](#), a crisis event occurs when monthly credit growth falls at or below its full-sample fourth percentile, provided no event occurred during the previous 36 months. Output paths show losses relative to normal growth and are normalized to zero in the month before the event. A crisis spell is a consecutive run of months with $C_t = 1$. A crisis episode begins in the event month and ends at the start of the first twelve consecutive months outside the crisis region. The twelve months used to confirm the end of the episode are excluded from its duration. [Reinhart and Rogoff \(2009\)](#) report an average peak-to-trough contraction in real per capita GDP of 1.9 years, or approximately 23 months, after severe financial crises. The empirical target for the persistence of the intermediation wedge is the recovery of credit spreads after financial crises in [Krishnamurthy and Muir \(2025\)](#), whose half-life is about 2.5 years ([Krishnamurthy and Li, 2025](#)).

Preferences and technology. Risk aversion is $\gamma = 10$ and the elasticity of intertemporal substitution is one. With unit EIS the consumption–wealth ratio equals ρ ; the discount rate of 8% and γ are set jointly to the unconditional risk premium and the risk-free rate. Depreciation is 5%, the aggregate rate implied by the BEA fixed-asset accounts and standard in this class of models. The

	Parameter	Choice
<i>Preferences & distribution</i>		
ψ	EIS	1.00
γ	Risk aversion	10
ρ	Discount rate	0.08
<i>Technology</i>		
ϕ	Investment friction	2.0
δ	Depreciation	0.05
σ	Capital volatility	0.07
a_e	Expert net payoff	0.17
a_h	Household net payoff	-0.05
<i>Capacity and exit</i>		
$\bar{m}$	Capacity - upper bound	0.20
$\underline{m}$	Capacity - lower bound	0.10
φ	Corr. of capacity and fundamental shock	0.25
κ_m^1, κ_m^2	Capacity mean reversion	0.001, 0.10
ξ_m	Capacity destroyed per unit exit	50
ν	Capacity volatility scale	25
β	Capacity volatility exponent	1.05
$\bar{\omega}$	Max. intermediation wedge	0.06
τ	Exit-opportunity rate	0.03
$\bar{\lambda}$	Switching cost	0.60
<i>Constraints</i>		
$\bar{\sigma}$	Tightness of the leverage constraint	0.12
$\underline{\chi}$	Skin-in-the game constraint	0.45

Table 1. Model parameters.

investment technology $\Phi(\iota) = \log(1 + \phi\iota)/\phi$ with $\phi = 2$ fixes the elasticity of the capital price to investment; the investment–capital ratio then follows from depreciation and trend growth through $\Phi(\iota) = \delta + g$: a 5% depreciation rate and 2% growth require net investment of 7%, and with the adjustment cost the gross ratio is $\iota \approx 8\%$, within the 7–9% range for aggregate fixed capital (e.g. [He and Krishnamurthy, 2019](#)).⁹ Capital volatility is $\sigma = 7\%$, set to the volatility of aggregate capital returns; together with the misallocation component in crisis states it implies an output–growth volatility of 5.7%, above the 3.8% in the data (Table 2). The expert net payoff a_e pins down the output–capital ratio, and the payoff gap Δa governs the output cost of misallocation and hence the

⁹Higher I/K ratios sometimes reported in the literature reflect narrower capital concepts weighted toward equipment and intellectual property, whose depreciation rates are three to four times those of the structures that dominate the aggregate stock.

depth of the crisis output decline.

Exit. The share of institutions on the FDIC problem-bank list averaged 3.5% of all FDIC-insured institutions between 2004 and 2023 and peaked at 11.5% during the global financial crisis. I use this series to discipline the arrival rate of exit opportunities and set $\tau = 0.03$. The annual bank failure rate, which averaged 0.4%, disciplines the mean realized exit rate $\mathbb{E}[\tau G(z_t)]$. For a small positive exit gap, the uniform switching-cost distribution implies $G(z) = \frac{z}{\bar{\lambda}}$. The realized exit rate is therefore approximately $\tau G(z) = \frac{\tau}{\bar{\lambda}} z$. The problem-bank share disciplines τ , while the bank failure rate disciplines $\bar{\lambda}$. Given $\tau = 0.03$, I choose $\bar{\lambda}$ to generate an average exit rate of 0.1%, compared with an annual bank failure rate of 0.4%, which is an upper bound for exit in the model. Exit destroys a bank's relationship capital in the model, whereas many failed banks are acquired and their lending relationships survive. The rate of capacity loss depends jointly on the exit rate and the amount of capacity destroyed per exit, $\xi_m \tau G(z_t)$. I therefore calibrate ξ_m together with the exit parameters.

Capacity. I construct an index of intermediation capacity from bank franchise value and credit-intermediation employment over 1993Q4–2020Q1. Franchise value is measured as the market value of listed bank-holding-company equity in excess of book value, scaled by assets. Employment is measured for credit intermediation under NAICS 522. I detrend and standardize both series and use their first principal component, which explains 84% of their joint variation. I choose the sign of the principal component so that it is positively correlated with franchise value, and then scale it to $[0, 1]$. The index has a mean of 0.50, falls to about 0.2 in 2009, and recovers slowly. Because it is based on only two observables and franchise value also reflects discount rates, I use the index only to discipline the dispersion and persistence of capacity. I do not interpret it as a structural estimate of capacity or use it to construct asset-pricing targets.

I jointly choose ν , ξ_m , and κ_m^2 to match these features. Simulated normalized capacity state has a standard deviation of 0.38, compared with 0.31 in the data, and a half-life of 30 months, compared with 48 months. I set κ_m^1 near zero because its estimated value is not statistically different from zero, $\hat{\kappa}_m = 0.17$ with a standard error of 0.10. The parameter $\bar{\omega}$ determines the range of the intermediation wedge. I set $\varphi = 0.25$, between the estimated correlations of capacity with GDP growth, 0.29 in quarterly changes and 0.21 after removing AR(1) components. I set β slightly above one so that capacity risk vanishes faster than expert activity. The results are insensitive to this choice.

Constraints. The skin-in-the-game floor $\underline{\chi} = 0.45$ requires experts to retain at least 45% of the equity in the capital they operate; the benchmark of Section 4.3 uses 0.5, and Figure 5 varies it

between 0.2 and 1. The tightness of the leverage constraint, $\bar{\sigma} = 0.12$, caps the volatility of expert net worth at 1.7 times the fundamental volatility σ ; the constraint is slack when experts' exposure per unit of net worth is close to one and binds only in the constrained region, so that, together with χ , it governs the size of that region and hence the unconditional frequency of crisis states (8.6% of months) and mean leverage (Table 2).

Model moments. Table 2 reports moments from simulating the economy for 2,000 years over 1,000 Monte Carlo paths at a monthly time step.¹⁰ The model generates a risk premium of 5.5% together with a risk-free rate of 1.5%, a combination that is difficult to obtain in asset-pricing models (Mehra and Prescott, 1985), with an unconditional Sharpe ratio of 0.79. It matches trend output growth and the investment–capital ratio and produces a sharp contraction in crisis states. Intermediary leverage of 2.9 is at the lower end of the 3–4 range measured for the aggregate intermediary sector (He and Krishnamurthy, 2019; Adrian et al., 2014) and is countercyclical, as in the data.

Moment	Model	Data
<i>Panel A. Asset pricing moments</i>		
Risk premium (%)	5.5	6.3
Risk-free rate (%)	1.5	0.9
Sharpe Ratio	0.79	0.5
<i>Panel B. Macroeconomic moments</i>		
I/K (%)	7.8	7–9
Output growth (%)	2.0	2.0
Crisis-state output growth (annualized, %)	-4.0	—
Output growth vol (%)	5.7	3.8
<i>Panel C. Intermediary moments</i>		
Assets/Equity (A/E)	2.9	3–4
Corr(A/E, $\Delta \log Y$)	-0.37	-0.18
Mean exit rate (%)	0.1	0.4
<i>Panel D. Crises moments</i>		
Crisis duration (months)	21	23
Capacity half-life (months)	30	48
Post-crisis wedge half-life (months)	29	30

Table 2. Model-implied moments and empirical benchmarks. A/E is defined as κ_t/η_t .

¹⁰Appendix D presents the simulation details.

Persistence. The slow-moving capacity state is the source of persistence in the model. Figure 3 traces the economy around crisis events. Capacity is already below its mean a year before the event and declines into it, reaching its floor at the onset; the intermediation wedge peaks at 5.9% against an unconditional mean of 3.1%; and both revert with a half-life of 29 months, close to the roughly 30-month half-life with which credit spreads recover after financial crises (Krishnamurthy and Muir, 2025; Krishnamurthy and Li, 2025). More than half of crisis events are followed by a relapse into the crisis region within a year, and the probability of being in that region is still 20% one year after an event and 16% after three, against 8.6% unconditionally. A crisis episode lasts 21 months on average (median 14), close to the 1.9-year average duration of the output downturn after severe financial crises in Reinhart and Rogoff (2009). Output falls by about 4% relative to trend on impact; the misallocation component of this decline reverses within a year, and the remaining shortfall of about 2% is the permanent level effect of the capital shock at the event, as in any AK economy. Accordingly, I use crisis-state output growth to assess crisis amplification and the event-time paths of capacity and the intermediation wedge to assess persistence. In the stationary distribution, capacity has a quarterly autocorrelation of 0.93, implying a half-life of 30 months compared with 48 months in the data. Following a crisis, the median time required for capacity to return to its unconditional median is about three years.

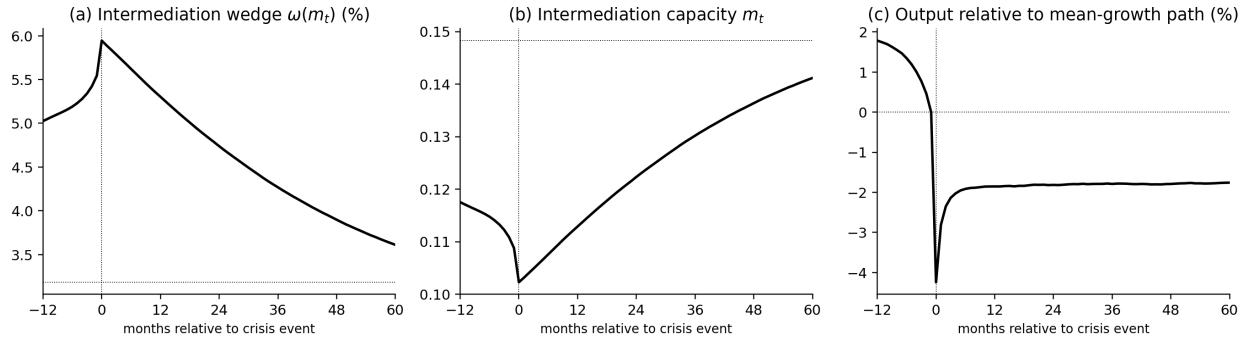


Figure 3. Event-time paths around crisis events. Month 0 denotes a fall in intermediated credit growth below its fourth percentile, with events separated by at least 36 months. Lines show event averages from months -12 to 60 . Panels (a)–(c) report the intermediation wedge, capacity, and output relative to its unconditional growth path. Dotted lines indicate unconditional means.

4.2 Counterfactual economies

To isolate the role of the channels in the exit–capacity mechanism, I solve and simulate four counterfactual economies, each of which imposes one restriction while holding all remaining parameters at the baseline calibration: $\xi_m = 0$ (experts exit, but exit does not destroy capacity), $\tau = 0$ (no expert exit), $m_t = \bar{m}$ (constant capacity: no capacity shocks and no erosion), and $\nu = 0$ (no direct capacity shocks). Table 3 reports the simulated moments alongside the baseline.

Moment	Baseline	$\xi_m=0$	$\tau=0$	$m_t=\bar{m}$	$\nu=0$
Risk premium (%)	5.5	5.1	4.9	5.0	5.6
Crisis-state output growth (annualized, %)	-4.0	—	—	—	—
Risk-free rate (%)	1.5	5.3	5.3	5.3	3.1
Leverage (A/E)	2.9	2.4	2.2	2.3	3.0
Intermediation wedge ω (%)	3.1	0.0	0.0	0.0	2.1
Mean capacity m_t	0.15	0.20	0.20	0.20	0.16
Mean wealth share η_t	0.34	0.42	0.45	0.44	0.34

Table 3. Counterfactual economies. Each column imposes one restriction on the exit–capacity mechanism, holding all remaining parameters at the baseline calibration. A dash indicates that the economy spends no time in the crisis region, so crisis-state moments are undefined. Crisis statistics are measured at the monthly simulation frequency.

Two results stand out. First, fragility arises from the interaction between exit and capacity loss rather than from exit alone. Shutting down exit ($\tau = 0$), eliminating its effect on capacity ($\xi_m = 0$), or fixing capacity at its upper bound ($m_t = \bar{m}$) produces nearly identical economies. In each case, capacity converges to its upper bound, the intermediation wedge vanishes, experts accumulate more net worth, and crises disappear.

The general-equilibrium adjustment is straightforward. Once exit can no longer erode capacity, capacity converges to its upper bound, the intermediation wedge disappears, and the mean expert wealth share rises from 0.34 to 0.42–0.45. A better-capitalized expert sector can operate the same capital stock with less leverage: 2.9 falls to 2.2–2.4. Experts consequently require less compensation for bearing aggregate risk. Relative to these counterfactuals, the baseline generates an additional 40–60 basis points of unconditional risk premium and a risk-free rate nearly four percentage points lower. The lower risk-free rate reflects greater precautionary demand for safe assets in an economy with a more levered intermediary sector exposed to crises.

The similarity of the three counterfactuals is informative. The mechanism requires exit to reduce capacity; removing either element breaks the feedback. The $\xi_m = 0$ economy makes this point most directly. Experts retain the option to exit, but exit no longer damages capacity, and the economy is nearly indistinguishable from the no-exit case. Exit and capacity loss therefore operate as mutually reinforcing components of a single mechanism. Second, capacity shocks trigger crises, while exit-induced erosion amplifies and propagates them. The economy with $\nu = 0$ illustrates this distinction. Exit continues to erode capacity, which remains below its upper bound and sustains a positive intermediation wedge. The expert wealth share and leverage remain close to their baseline values, and the risk premium is essentially unchanged at 5.6%, compared with 5.5% in the baseline. Nevertheless, the economy never enters the crisis region.

Without capacity shocks, erosion is self-limiting. The capacity drift in equation (14) converges to a stable steady state at which rebuilding offsets erosion. Capacity therefore remains near 0.16,

compared with a baseline mean of 0.15, and does not fall sufficiently to generate a crisis. The risk-free rate is 3.1%, between 1.5% in the baseline and 5.3% in the absence of the mechanism. Exit-induced erosion alone thus accounts for slightly more than half of the decline in the baseline risk-free rate, while the crisis states generated by capacity shocks account for the remainder. Capacity shocks provide the initial displacement, and endogenous exit governs its propagation by determining whether capacity recovers or deteriorates further.

Taken together, the counterfactuals assign complementary roles to the two components of the mechanism. The interaction between exit and capacity erosion generates fragility, raises the risk premium, and accounts for most of the decline in the risk-free rate. Capacity shocks determine when crises occur. The next subsection studies a benchmark with no exit and constant capacity. The benchmark reproduces the amplification present in standard intermediary models but not the persistent capacity impairment generated by the full model.

4.3 Benchmark model

In this section, I study a benchmark that shuts down the exit–capacity channel. I set the exit-opportunity rate to zero, $\tau = 0$, eliminating both the exit hazard $\tau G(z_t)$ and the exit option in the expert’s problem. I also set $\nu = 0$ and initialize capacity at $m_t = \bar{m}$. Capacity then remains at its upper bound, the shock Z^m is absent, and the intermediation wedge satisfies $\omega(\bar{m}) = 0$. The expert wealth share η_t is the only state variable, and the economy reduces to Brunnermeier and Sannikov (2014) with recursive preferences and outside equity.

I calibrate the benchmark separately using values standard in this class of models.¹¹ The comparison is therefore between two separately calibrated economies rather than a marginal counterfactual. Table 3 reports the corresponding marginal counterfactuals.

Without exit, the benchmark has a degenerate long-run distribution. The expert wealth share converges to the skin-in-the-game floor $\underline{\chi}$, where experts and households have identical exposures. Both the drift and volatility of η_t then vanish, and the economy no longer returns to the crisis region, as shown in the $\tau = 0$ column of Table 3. To obtain a nondegenerate stationary distribution with recurrent crises, I introduce overlapping-generations turnover at rate $\lambda_d > 0$, following Gârleanu and Panageas (2015). Turnover mean-reverts the expert net worth and prevents long-run distribution from becoming degenerate.

The benchmark generates the nonlinear amplification familiar from intermediary models. Negative shocks reduce output and investment and raise the risk premium. It cannot, however, jointly

¹¹The benchmark parameters are $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$, $\underline{\chi} = 0.5$, $\phi = 10$, $\delta = 0.02$, $\sigma = 0.06$, and $\lambda_d = 0.03$. Departing wealth is reassigned so that experts receive a share $\bar{\eta} = 0.10$ of newborn wealth, as described in Appendix E. The leverage constraint is omitted. Risk aversion varies across experiments, with $RA = 1$ denoting the log-utility limit. Panel (b) of Figure 4 sets $a_e = 0.05$ and $a_h = -0.03$, leaving the remaining parameters unchanged. Unlike the full model, the benchmark includes overlapping-generations turnover.

reproduce the observed unconditional risk premium, crisis amplification, and persistent impairment. With $RA = 1$, annualized output growth averages -10% in crisis states. The wealth share remains in the crisis region for about five months, while episodes that include relapses last 12 months on average and 2 months at the median, compared with 21 months in the full model. The mean is driven by a small number of severe wealth-share collapses. Despite this amplification, the unconditional risk premium is only about 2% .

Any post-crisis persistence in the benchmark comes solely from the gradual recovery of expert net worth and is quantitatively relevant only near $RA = 1$. Raising risk aversion toward the level required to match the unconditional risk premium causes the crisis probability to approach zero. The dates selected by the credit-based event rule then rarely coincide with crisis states. Figure 4 summarizes this limitation: panel (a) relates output amplification to crisis duration as risk aversion varies, while panel (b) shows how the crisis probability declines as the unconditional risk premium rises.

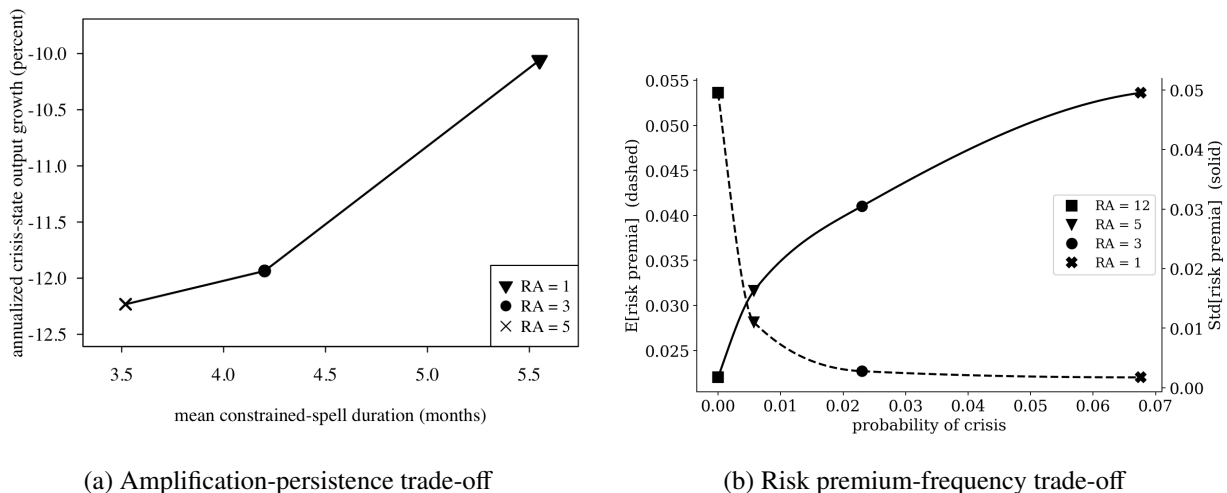


Figure 4. (a) The left panel plots annualized crisis-state output growth against the mean constrained-spell duration (consecutive months in the crisis region) implied by the benchmark model for three levels of risk aversion. (b) The right panel plots the unconditional risk premium on the left axis and the standard deviation of the risk premium on the right axis against the probability of crisis for different risk aversion levels.

Increasing risk aversion strengthens amplification but shortens crises. The benchmark is driven only by Brownian innovations, which introduce no exogenous persistence. Recovery therefore depends on the endogenous drift of the expert wealth share. When the economy enters a crisis, the risk premium rises sharply. Surviving experts earn high expected returns on their levered positions and rapidly rebuild net worth. Higher risk aversion raises the crisis risk premium further, increasing the recapitalization drift and accelerating the return to the normal region. The same force that amplifies the initial response therefore reduces its persistence. Appendix Figure 6 illustrates this mechanism over the broader range $RA \in \{1, 5, 12\}$.

Panel (b) of Figure 4 shows a related tension between the unconditional risk premium and the

incidence of crises. With $RA = 1$, the benchmark spends 6.7% of the time in the crisis region but generates an unconditional risk premium of only about 2%. Raising risk aversion increases the unconditional premium, as is standard in asset-pricing models, but also raises experts' expected returns and shifts the stationary distribution of their wealth share away from the crisis region. Crises consequently become less frequent. These results are robust to recursive preferences because the risk-premium channel dominates the effects of the consumption–wealth ratio and investment on the expected growth of expert net worth. The same trade-offs arise when crises are dated using the credit rule in Section 4.1.

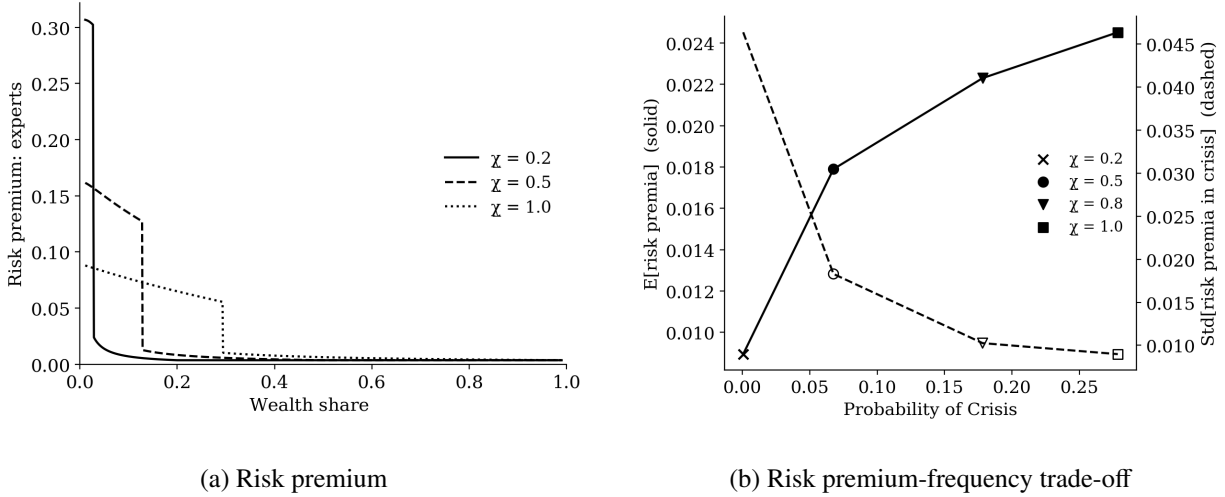


Figure 5. (a) The left panel presents risk premium on the y-axis and wealth share of experts on the x-axis for three different skin-in-the-game constraint parameters. (b) The right panel plots the unconditional risk premium on the left axis and the standard deviation of the risk premium on the right axis against probability of crisis for different skin-in-the-game constraint parameters.

A central feature of the benchmark is its restriction on outside equity. The parameter χ is the minimum equity stake experts must retain and therefore governs the tightness of the skin-in-the-game constraint. Tightening this constraint shifts the crisis boundary to the right and increases the probability of entering the crisis region, as shown in panel (a) of Figure 5. It has little effect, however, on the unconditional risk premium, as shown in panel (b).

Appendix Figure 7 examines whether varying the OLG turnover parameters can overcome this limitation. Generating a crisis probability of approximately 7% requires an expert population share below 10% and a death rate of about 3%. Because death risk enters the effective discount rate, a death rate of this magnitude leaves almost no scope for pure time discounting. Even under these parameter values, the unconditional risk premium is only 2.4%, well below the empirical target of 6.3%. Adjusting the turnover parameters can therefore increase the frequency of crises, but it does not resolve the benchmark's risk-premium shortfall. Nor does it generate persistent impairment: constrained episodes remain short, and no state variable carries the effects of a crisis forward once

expert net worth has recovered.

5 Conclusion

Financial crises combine sharp amplification with slow recovery. Standard intermediary models generate amplification through balance-sheet constraints, but the resulting increase in risk premia also allows surviving intermediaries to rebuild net worth, limiting persistence. This paper introduces an exit–capacity feedback. When remaining in the intermediary sector becomes sufficiently unattractive, experts exit and reduce the sector’s intermediation capacity. Lower capacity further reduces the value of remaining an expert and induces additional exit, while gradual rebuilding transforms temporary balance-sheet distress into persistent capacity impairment.

The boundary analysis characterizes the race between recapitalization and exit. Combining the resulting exit behavior with capacity dynamics can produce locally trapping and locally escaping regimes when the feedback is sufficiently strong. The calibrated economy has a stationary distribution and generates realistic crisis amplification, endogenous exit, and persistent capacity impairment. By contrast, the benchmark without exit and capacity loss cannot jointly produce recurrent crises, high risk premia, and persistent damage: parameter changes that raise the risk premium drive the crisis probability toward zero. Persistent losses of intermediation capacity can therefore sustain slow recoveries even when the economy ultimately returns to normal.

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A Derivations

Throughout, the two shocks (Z^k, Z^m) have correlation φ , so for loadings $\mathbf{a} = (a^k, a^m)$, $\mathbf{b} = (b^k, b^m)$ I write $\langle \mathbf{a}, \mathbf{b} \rangle = a^k b^k + a^m b^m + \varphi(a^k b^m + a^m b^k)$ and $\|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle$. The capital-return loading is $\sigma_R = (\sigma + \sigma^{q,k}, \sigma^{q,m})$, so $\|\sigma_R\|^2 = (\sigma + \sigma^{q,k})^2 + (\sigma^{q,m})^2 + 2\varphi(\sigma + \sigma^{q,k})\sigma^{q,m}$. To distinguish physical capital operation from financial ownership, let

$$K_{e,t} = \kappa_t K_t$$

denote physical capital operated by experts, and let

$$K_{h,t}^{\text{op}} = (1 - \kappa_t) K_t$$

denote physical capital operated directly by households. Physical capital market clearing is therefore

$$K_{e,t} + K_{h,t}^{\text{op}} = K_t.$$

Households also hold the outside equity issued against expert-operated capital. Let

$$E_{h,t}^{\text{out}} = (1 - \chi_t) q_t K_{e,t}$$

denote the market value of these outside-equity claims. Total household risky financial exposure is

$$X_{h,t} := q_t K_{h,t}^{\text{op}} + E_{h,t}^{\text{out}}.$$

Using $K_{e,t} = \kappa_t K_t$ gives

$$X_{h,t} = (1 - \chi_t \kappa_t) q_t K_t.$$

Thus, $K_{h,t}^{\text{op}}$ denotes household-operated physical capital, whereas $X_{h,t}$ denotes the household sector's total risky financial exposure, including outside equity. Let $k_{h,t}^{\text{op}} \geq 0$ denote the physical capital operated directly by a household, and let $e_{h,t}^{\text{out}} \geq 0$ denote the market value of the household's outside-equity claims. Define its total risky financial exposure by

$$x_{h,t} := q_t k_{h,t}^{\text{op}} + e_{h,t}^{\text{out}}.$$

The problem of households is

$$\max_{c_{h,t}, k_{h,t}^{\text{op}}, e_{h,t}^{\text{out}}, l_{h,t}} U_{h,t} := \mathbb{E}_t \left[\int_t^\infty f(c_s, U_s) ds \right]$$

subject to the wealth constraint given by

$$\begin{aligned} \frac{dw_{h,t}}{w_{h,t}} = & \left[r_t - \frac{c_{h,t}}{w_{h,t}} + \frac{q_t k_{h,t}^{\text{op}}}{w_{h,t}} (\mu_{R,h,t} - r_t) + \frac{e_{h,t}^{\text{out}}}{w_{h,t}} \epsilon_{h,t} + \mathcal{T}_{h,t} \right] dt \\ & + \frac{x_{h,t}}{w_{h,t}} \left(\sigma + \sigma_t^{q,k} \right) dZ_t^k + \frac{x_{h,t}}{w_{h,t}} \sigma_t^{q,m} dZ_t^m \end{aligned} \quad (34)$$

where

$$\epsilon_{h,t} := \langle \mathbf{s}_{h,t}, \boldsymbol{\sigma}_{R,t} \rangle$$

is the risk compensation required by households per unit of risky exposure, and the aggregator $f(c, U)$ is given in (6). The actual excess return on household-operated physical capital satisfies

$$\mu_{R,h,t} - r_t \leq \epsilon_{h,t},$$

with equality whenever households operate a positive quantity of physical capital. Equivalently,

$$0 \leq k_{h,t}^{\text{op}} \quad \perp \quad \epsilon_{h,t} - (\mu_{R,h,t} - r_t) \geq 0.$$

and $\mathcal{T}_{h,t} = \omega(m_t)\kappa_t/(1 - \eta_t)$ is the intermediation wedge rebated to households per unit of their wealth. They face a no-shorting constraint (i.e., $k_{h,t}^{\text{op}} \geq 0$, $e_{h,t}^{\text{out}} \geq 0$).

Definition 2 (Equilibrium). *An equilibrium is a set of price processes $\{q_t, r_t\}$, allocations $\{c_{j,t}, \iota_{j,t}\}_{j \in \{e,h\}}$, $\{k_{e,t}, \chi_t\}$, $\{k_{h,t}^{\text{op}}, e_{h,t}^{\text{out}}\}$, and exit decisions such that the following conditions hold:*

1. *Given prices, experts solve the recursive stopping problem (7) subject to the wealth dynamics (8), the outside-equity constraint $\chi_t \geq \chi$, and the leverage constraint (13b); households solve their problem subject to (34) and the no-shorting constraints (Appendix A).*

2. *Physical capital clears:*

$$K_{e,t} + K_{h,t}^{\text{op}} = K_t.$$

3. *Outside-equity claims clear:*

$$E_{h,t}^{\text{out}} = (1 - \chi_t)q_t K_{e,t}.$$

4. *The goods market clears:*

$$C_{e,t} + C_{h,t} = (a_e - \iota_t)K_{e,t} + (a_h - \iota_t)K_{h,t}^{\text{op}}.$$

5. *The bond market clears by Walras' law.*

I solve for a Markov equilibrium in two state variables, $(\eta_t, m_t) \in \Omega := (0, 1) \times [\underline{m}, \overline{m}]$.

Definition 3. *A Markov equilibrium in Ω is a set of adapted processes $q(\eta_t, m_t)$, $r(\eta_t, m_t)$, $J_e(\eta_t, m_t)$, $J_h(\eta_t, m_t)$, policy functions $c_e(\eta_t, m_t)$, $c_h(\eta_t, m_t)$, $\kappa(\eta_t, m_t)$, $\chi(\eta_t, m_t)$, $\iota(\eta_t, m_t)$, and state variables (η_t, m_t) such that for $j \in \{e, h\}$*

- *J_j solves the HJB equation of type j (equations (35)–(36)) and the policy functions attain the maximum, with the exit decision given by the threshold rule on z_t ;*
- *the pricing conditions (13)–(13b) and the household capital-allocation complementarity condition hold;*
- *the goods market clears,*

$$[C_{e,t}/W_{e,t}\eta_t + C_{h,t}/W_{h,t}(1 - \eta_t)] q_t = \kappa_t(a_e - \iota_t) + (1 - \kappa_t)(a_h - \iota_t),$$

and capital and outside equity clear as in the definition above;

- η_t and m_t satisfy (18)-(20) and (14)

where the capital letters denote aggregate quantities. Let $\mathcal{J}_j(\eta, m)$ denote the wealth-scaled multiplier in an individual value function, and define the sector-capital-scaled multipliers $J_e := \eta q \mathcal{J}_e$ and $J_h := (1 - \eta) q \mathcal{J}_h$. Hence, $KJ_j = W_j \mathcal{J}_j$. For an incumbent individual i , however, $w_{e,i} \mathcal{J}_e = s_{e,i} KJ_e$ and $w_{h,i} \mathcal{J}_h = s_{h,i} KJ_h$, where $s_{j,i} := w_{j,i}/W_j$. Exit reclassifies aggregate wealth across sectors without changing the wealth of a continuing incumbent. Thus, with $x_t := \tau G(z_t)$,

$$\left. \frac{ds_{e,i,t}}{s_{e,i,t}} \right|_{\text{exit}} = x_t dt, \quad \left. \frac{ds_{h,i,t}}{s_{h,i,t}} \right|_{\text{exit}} = -\frac{\eta_t}{1 - \eta_t} x_t dt.$$

The common state generator determines the drift of the sector-scaled multipliers, while the individual HJBs use the corrected drifts defined below. The exit gap remains

$$z_t = 1 - \frac{\mathcal{J}_{e,t}}{\mathcal{J}_{h,t}} = 1 - \frac{(1 - \eta_t) J_{e,t}}{\eta_t J_{h,t}}.$$

The HJB equation of experts is given by

$$\begin{aligned} 0 = & \rho [\log(\rho \eta q_t) - \log J_{e,t}] + (\Phi(\iota_t) - \delta) - \frac{\gamma}{2} \sigma^2 + \hat{\mu}_{e,t}^J - \frac{\gamma}{2} \left((\sigma_{e,t}^{J,k})^2 + (\sigma_{e,t}^{J,m})^2 + 2\varphi \sigma_{e,t}^{J,k} \sigma_{e,t}^{J,m} \right) \\ & + (1 - \gamma) \left(\sigma \sigma_{e,t}^{J,k} + \varphi \sigma \sigma_{e,t}^{J,m} \right) + \frac{\tau}{(1 - \gamma) \bar{\lambda}} \int_0^{\min\{z_t^+, \bar{\lambda}\}} \left[\left(\frac{(1 - \lambda) \eta_t J_{h,t}}{(1 - \eta_t) J_{e,t}} \right)^{1-\gamma} - 1 \right] d\lambda \end{aligned} \quad (35)$$

The HJB equation of the household is

$$\begin{aligned} 0 = & \rho [\log(\rho(1 - \eta) q_t) - \log J_{h,t}] - \frac{\gamma}{2} \sigma^2 + \hat{\mu}_{h,t}^J - \frac{\gamma}{2} \left((\sigma_{h,t}^{J,k})^2 + (\sigma_{h,t}^{J,m})^2 + 2\varphi \sigma_{h,t}^{J,k} \sigma_{h,t}^{J,m} \right) \\ & + (\Phi(\iota_t) - \delta) + (1 - \gamma) \left(\sigma \sigma_{h,t}^{J,k} + \varphi \sigma \sigma_{h,t}^{J,m} \right) \end{aligned} \quad (36)$$

The drift of opportunity processes is obtained using Ito's lemma.

$$\begin{aligned} \mu_{j,t}^{J,\text{sec}} J_{j,t} = & \frac{\partial J_{j,t}}{\partial \eta_t} \mu_t^\eta \eta_t + \frac{\partial J_{j,t}}{\partial m_t} \mu_{m,t} + \frac{1}{2} \frac{\partial^2 J_{j,t}}{\partial \eta_t^2} \eta_t^2 \left[(\sigma_t^{\eta,k})^2 + (\sigma_t^{\eta,m})^2 + 2\varphi \sigma_t^{\eta,k} \sigma_t^{\eta,m} \right] \\ & + \frac{\partial^2 J_{j,t}}{\partial \eta_t \partial m_t} \sigma_{m,t} \left[\eta_t \sigma_t^{\eta,m} + \varphi \eta_t \sigma_t^{\eta,k} \right] + \frac{1}{2} \frac{\partial^2 J_{j,t}}{\partial m_t^2} \sigma_{m,t}^2, \end{aligned} \quad (37a)$$

$$\hat{\mu}_{e,t}^J = \mu_{e,t}^{J,\text{sec}} + x_t, \quad \hat{\mu}_{h,t}^J = \mu_{h,t}^{J,\text{sec}} - \frac{\eta_t}{1 - \eta_t} x_t, \quad x_t = \tau G(z_t). \quad (37b)$$

Similarly, diffusion functions are given by

$$\sigma_{j,t}^{J,k} J_{j,t} = \frac{\partial J_{j,t}}{\partial \eta_t} \eta_t \sigma_t^{\eta,k}; \quad \sigma_{j,t}^{J,m} J_{j,t} = \frac{\partial J_{j,t}}{\partial \eta_t} \eta_t \sigma_t^{\eta,m} + \frac{\partial J_{j,t}}{\partial m_t} \sigma_{m,t} \quad (38)$$

A.1 Consumption and investment

With unit IES the Duffie–Epstein aggregator yields the consumption–wealth ratio $c_{j,t}/w_{j,t} = \rho$ for both types. All agents choose the investment rate from $\max_t [(a_j - \iota)/q + \Phi(\iota)]$; since $\Phi'(\iota) = 1/(\phi\iota + 1)$, the first-order condition $\Phi'(\iota) = 1/q$ gives the common rate $\iota_t = (q_t - 1)/\phi$.

A.2 Capital price

From the goods market clearing, $C_{e,t} + C_{h,t} = (a_e - \iota_t)K_{e,t} + (a_h - \iota_t)K_{h,t}^{\text{op}} = (\bar{a}_t - \iota_t)K_t$ with $\bar{a}_t := \kappa_t a_e + (1 - \kappa_t)a_h$, together with $C_e + C_h = \rho(W_e + W_h) = \rho q_t K_t$ and $\iota_t = (q_t - 1)/\phi$, gives

$$\rho q_t = \bar{a}_t - \frac{q_t - 1}{\phi} \implies q_t = \frac{1 + \phi \bar{a}_t}{1 + \phi \rho} \quad (39)$$

Note that as $\kappa_t \rightarrow 0$, $q_t \rightarrow (1 + \phi a_h)/(1 + \phi \rho) = q_0$.

A.3 Prices of risk and the pricing conditions (13)–(17)

The value function guess is $U_j = (w_j \mathcal{J}_j)^{1-\gamma}/(1-\gamma)$, and the marginal value of wealth is proportional to $w_j^{-\gamma} \mathcal{J}_j^{1-\gamma}$, so the price of risk loads on the *wealth-scaled* multiplier $\mathcal{J}_j$ (related to the HJB's J_j by $J_j = (\text{share})_q \mathcal{J}_j$). Hence, the equilibrium price-of-risk vector of agent j is

$$\boldsymbol{\varsigma}_{j,t} = \gamma \boldsymbol{\sigma}_{j,t}^w + (\gamma - 1) \boldsymbol{\sigma}_{j,t}^{\mathcal{J}},$$

and the risk premium it requires per unit of capital exposure is $\epsilon_{j,t} := \langle \boldsymbol{\varsigma}_{j,t}, \boldsymbol{\sigma}_R \rangle$. The wealth diffusion loadings are $\boldsymbol{\sigma}_e^w = \frac{\chi^K}{\eta} \boldsymbol{\sigma}_R$ for experts and $\boldsymbol{\sigma}_h^w = \frac{1-\chi^K}{1-\eta} \boldsymbol{\sigma}_R$ for households; the value-multiplier loadings, from Itô's lemma on $\mathcal{J}_j(\eta, m)$, are

$$\boldsymbol{\sigma}_j^{\mathcal{J},k} = \frac{\mathcal{J}_{j,\eta} \eta}{\mathcal{J}_j} \boldsymbol{\sigma}^{\eta,k}, \quad \boldsymbol{\sigma}_j^{\mathcal{J},m} = \frac{\mathcal{J}_{j,\eta} \eta}{\mathcal{J}_j} \boldsymbol{\sigma}^{\eta,m} + \frac{\mathcal{J}_{j,m}}{\mathcal{J}_j} \boldsymbol{\sigma}_m.$$

Remark: The boundary limits are: as $\eta \rightarrow 0$, $\boldsymbol{\sigma}_h^w \rightarrow (\boldsymbol{\sigma}, 0)$ and $\boldsymbol{\sigma}_h^{\mathcal{J}} \rightarrow 0$, so $\epsilon_h \rightarrow \gamma \sigma^2$.

A.4 Price dynamics: amplification

Since $q = q(\eta, m)$ with $d\eta = \eta(\mu^\eta dt + \sigma^{\eta,k} dZ^k + \sigma^{\eta,m} dZ^m)$ and $dm = \mu_m dt + \sigma_m dZ^m$, Itô's lemma gives the price diffusions

$$\sigma^{q,k} = \frac{q_\eta}{q} \eta \sigma^{\eta,k} \quad \sigma^{q,m} = \frac{q_\eta}{q} \eta \sigma^{\eta,m} + \frac{q_m}{q} \sigma_m$$

Substituting out above equations and writing the amplification coefficient $A := \frac{q_\eta \eta}{q} \left(\frac{\chi^K}{\eta} - 1 \right)$,

$$\sigma^{q,k} = \frac{A \sigma}{1 - A}, \quad \sigma^{q,m} = \frac{(q_m/q) \sigma_m}{1 - A} \quad (40)$$

the loss-spiral amplification $1/(1 - A)$ similar to Brunnermeier and Sannikov (2014), augmented by a *capacity* channel that enters through q_m . The price drift, from the Itô drift of $q(\eta, m)$, is

$$\mu_q = \frac{1}{q} \left[q_\eta \eta \mu^\eta + q_m \mu_m + \frac{1}{2} q_{\eta\eta} \eta^2 \|\sigma^\eta\|^2 + \frac{1}{2} q_{mm} \sigma_m^2 + q_{\eta m} \eta \sigma_m (\sigma^{\eta,m} + \varphi \sigma^{\eta,k}) \right] \quad (41)$$

with μ^η from (18), μ_m from (14), and $\|\sigma^\eta\|^2 = (\sigma^{\eta,k})^2 + (\sigma^{\eta,m})^2 + 2\varphi \sigma^{\eta,k} \sigma^{\eta,m}$.

B Proofs

B.1 Proof of equations (18)–(20).

Wealth share of expert is denoted as $\eta_t = W_{e,t}/N_t$ with $N_t := q_t K_t$ the aggregate net worth. Total wealth in the economy is $W_{e,t} + W_{h,t} = N_t$. Recall the individual wealth dynamics¹²,

$$\frac{dw_e}{w_e} = \mu_e^w dt + \chi \frac{qk_e}{w_e} (\sigma + \sigma^{q,k}) dZ^k + \chi \frac{qk_e}{w_e} \sigma^{q,m} dZ^m, \quad \mu_e^w = r - \frac{c_e}{w_e} + \frac{qk_e}{w_e} (\tilde{\mu}_{R,e} - r) - (1 - \chi) \frac{qk_e}{w_e} \epsilon_h,$$

with $\tilde{\mu}_{R,e} = \mu_{R,e} - \omega(m)$. For households, let k_h^{op} denote directly operated physical capital and let e_h^{out} denote outside-equity holdings. Their total risky financial exposure is

$$x_h = qk_h^{\text{op}} + e_h^{\text{out}}.$$

The household wealth drift is

$$\mu_h^w = r - \frac{c_h}{w_h} + \frac{qk_h^{\text{op}}}{w_h} (\mu_{R,h} - r) + \frac{e_h^{\text{out}}}{w_h} \epsilon_h + \mathcal{T}_h,$$

where

$$\epsilon_h = \langle \mathbf{S}_h, \boldsymbol{\sigma}_R \rangle.$$

The household capital-pricing condition implies

$$\mu_{R,h} - r \leq \epsilon_h,$$

with equality whenever $k_h^{\text{op}} > 0$. Expert-operated physical capital is $K_e = \kappa K$ while household-operated physical capital is $K_h^{\text{op}} = (1 - \kappa)K$. Experts issue outside equity with aggregate market value $E_h^{\text{out}} = (1 - \chi)qK_e$. Household total risky financial exposure is therefore

$$X_h = qK_h^{\text{op}} + E_h^{\text{out}} = (1 - \chi\kappa)qK.$$

Since $N = qK$ and $W_h = (1 - \eta)N$,

$$\frac{X_h}{W_h} = \frac{1 - \chi\kappa}{1 - \eta}.$$

If households operate positive physical capital, their capital-pricing condition holds with equal-

¹²Ignoring time index for simplicity

ity. If they operate no physical capital, $qK_h^{\text{op}} = 0$. Hence, in either regime, the aggregate household excess-return income can be written as

$$\frac{X_h}{W_h} \epsilon_h.$$

Diffusions. From the expert wealth dynamics, $\sigma_{W_e} = \frac{\chi\kappa}{\eta} \sigma_R$. Since $N = qK$, $dN/N = dq/q + dK/K + \langle dq/q, dK/K \rangle$ has diffusion $\sigma_N = (\sigma + \sigma^{q,k}, \sigma^{q,m}) = \sigma_R$. By applying the Ito quotient rule,

$$\mu^\eta = \mu_{W_e} - \mu_N + \|\sigma_N\|^2 - \langle \sigma_{W_e}, \sigma_N \rangle, \quad \sigma^\eta = \sigma_{W_e} - \sigma_N.$$

The diffusion gives $\sigma^\eta = (\frac{\chi\kappa}{\eta} - 1) \sigma_R$, i.e. $\sigma^{\eta,k} = (\frac{\chi\kappa}{\eta} - 1)(\sigma + \sigma^{q,k})$ and $\sigma^{\eta,m} = (\frac{\chi\kappa}{\eta} - 1) \sigma^{q,m}$.

The covariation terms give $\|\sigma_N\|^2 - \langle \sigma_{W_e}, \sigma_N \rangle = (1 - \frac{\chi\kappa}{\eta}) \|\sigma_R\|^2 = -(\frac{\chi\kappa}{\eta} - 1) \|\sigma_R\|^2$.

Drift. By the law of large numbers, exit removes departing experts' total wealth at a rate $\tau G(z)$, so $\mu_{W_e} = \mu_e^w - \tau G(z)$. Hence $\mu_N = \eta \mu_e^w + (1 - \eta) \mu_h^w$. Hence

$$\mu_{W_e} - \mu_N = (1 - \eta)(\mu_e^w - \mu_h^w) - \tau G(z).$$

Using $c_j/w_j = \rho$ and the aggregate ratios above,

$$\mu_e^w - \mu_h^w = \frac{\kappa}{\eta} (\tilde{\mu}_{R,e} - r) - (1 - \chi) \frac{\kappa}{\eta} \epsilon_h - \frac{1 - \chi\kappa}{1 - \eta} \epsilon_h - \frac{\omega(m)\kappa}{1 - \eta},$$

so that

$$(1 - \eta)(\mu_e^w - \mu_h^w) = \frac{(1 - \eta)\kappa}{\eta} [(\tilde{\mu}_{R,e} - r) - (1 - \chi)\epsilon_h] - (1 - \chi\kappa)\epsilon_h - \omega(m)\kappa.$$

Collecting drift and covariation,

$$\mu^\eta = \frac{(1 - \eta)\kappa}{\eta} [(\tilde{\mu}_{R,e} - r) - (1 - \chi)\epsilon_h] - (1 - \chi\kappa)\epsilon_h - \omega(m)\kappa - \left(\frac{\chi\kappa}{\eta} - 1\right) \|\sigma_R\|^2 - \tau G(z),$$

which is equation (18). ■

B.2 Proof of Lemma 1.

Fix capacity at $m = m_0$. I first derive the limiting wealth-share dynamics and then characterize exit at the boundary.

Boundary drift and diffusion. Take the limit $\eta \downarrow 0$ in equations (18)–(20). Under Assumption 1,

$$\chi_0 = \underline{\chi}, \quad \frac{\chi_0 \kappa}{\eta} \longrightarrow \frac{\bar{\sigma}}{\sigma}, \quad \frac{\kappa}{\eta} \longrightarrow \ell_0 := \frac{\bar{\sigma}}{\underline{\chi} \sigma}.$$

In particular, $\kappa = \ell_0 \eta + o(\eta)$, so $\kappa \rightarrow 0$. From equation (39),

$$q \longrightarrow q_0 := \frac{1 + \phi a_h}{1 + \phi \rho}.$$

Moreover, bounded derivatives of q , together with

$$\sigma_m(\eta, m) = \nu \kappa^\beta (\bar{m} - m)(m - \underline{m}), \quad \beta > 1,$$

imply

$$\sigma^{q,k} \longrightarrow 0, \quad \sigma^{q,m} \longrightarrow 0.$$

Hence

$$\sigma_R \longrightarrow (\sigma, 0), \quad \|\sigma_R\|^2 \longrightarrow \sigma^2.$$

Equations (19) and (20) then give

$$(\sigma^{\eta,k}, \sigma^{\eta,m}) \longrightarrow \left(\left(\frac{\bar{\sigma}}{\sigma} - 1 \right) \sigma, 0 \right) = (\bar{\sigma} - \sigma, 0).$$

Defining

$$\sigma_0 := \bar{\sigma} - \sigma,$$

the limiting scalar diffusion coefficient is therefore

$$\lim_{\eta \downarrow 0} \sigma^\eta(\eta, m_0) = \sigma_0.$$

For households, aggregate risky exposure converges to one and the volatility of the household value multiplier vanishes. Thus,

$$\sigma_{h,0}^w = (\sigma, 0), \quad \sigma_{h,0}^J = (0, 0),$$

and

$$\varsigma_{h,0} = \gamma(\sigma, 0), \quad \epsilon_{h,0} = \langle \varsigma_{h,0}, \sigma_{R,0} \rangle = \gamma\sigma^2.$$

From the expert return relation $\tilde{\mu}_{R,e} - r = \epsilon_h + \Delta a/q - \omega(m)$ derived in Section 2

$$\tilde{\mu}_{R,e} - r = \epsilon_h + \frac{\Delta a}{q} - \omega(m).$$

Define

$$\pi(m_0) := \frac{\Delta a}{q_0} - \omega(m_0), \quad x_0(m_0) := \tau G(z_0(m_0)).$$

Taking the limit in equation (18), and using $\omega(m_0)\kappa \rightarrow 0$, gives

$$\begin{aligned} \mu_0(m_0) &:= \lim_{\eta \downarrow 0} \mu^\eta(\eta, m_0) \\ &= \ell_0 \pi(m_0) + (\chi_0 \ell_0 - 1) (\epsilon_{h,0} - \sigma^2) - x_0(m_0). \end{aligned} \tag{42}$$

Since

$$\chi_0 \ell_0 - 1 = \frac{\bar{\sigma}}{\sigma} - 1 = \frac{\sigma_0}{\sigma}$$

and $\epsilon_{h,0} = \gamma\sigma^2$, this becomes

$$\mu_0(m_0) = \ell_0\pi(m_0) + (\gamma - 1)\sigma\sigma_0 - x_0(m_0). \quad (43)$$

Boundary value gap and exit. All value multipliers in what follows are wealth-scaled. Define

$$\Delta_0(m_0) := \log \frac{\mathcal{J}_{e,0}(m_0)}{\mathcal{J}_{h,0}(m_0)}, \quad z_0(m_0) := 1 - e^{\Delta_0(m_0)}.$$

For an arbitrary candidate value gap Δ , let

$$\widehat{z}(\Delta) := \min \left\{ (1 - e^\Delta)^+, \bar{\lambda} \right\},$$

and define the value of the expert's exit option by

$$\mathcal{V}(\Delta) := \frac{\tau}{(1 - \gamma)\bar{\lambda}} \int_0^{\widehat{z}(\Delta)} \left[\left((1 - \lambda)e^{-\Delta} \right)^{1-\gamma} - 1 \right] d\lambda. \quad (44)$$

Because capacity and exposure are held fixed in the restricted continuation problem, the boundary HJBs are algebraic. Specializing the sector HJBs (35)–(36) to the maintained policy in Assumption 1 gives the two conditions below, in which the $\gamma\sigma\sigma_0$ term is the covariance of expert wealth with the capital return at exposure $\bar{\sigma}/\sigma$. Let

$$\Phi_0 := \frac{\log q_0}{\phi}.$$

The household boundary HJB is

$$\rho \log \mathcal{J}_{h,0} = \rho \log \rho + \Phi_0 - \delta - \frac{\gamma}{2}\sigma^2, \quad (45)$$

while the expert boundary HJB is

$$\rho \log \mathcal{J}_{e,0} = \rho \log \rho + \Phi_0 - \delta + \ell_0\pi(m_0) + \gamma\sigma\sigma_0 - \frac{\gamma}{2}\bar{\sigma}^2 + \mathcal{V}(\Delta_0(m_0)). \quad (46)$$

Subtracting equation (45) from equation (46), and using $\bar{\sigma} = \sigma + \sigma_0$, yields the exact boundary value-gap equation

$$\rho\Delta_0(m_0) = \ell_0\pi(m_0) - \frac{\gamma}{2}\sigma_0^2 + \mathcal{V}(\Delta_0(m_0)). \quad (47)$$

I now obtain the closed-form approximation used in the main text. In the small interior-exit region,

$$0 < z_0(m_0) < \bar{\lambda},$$

so $\widehat{z}(\Delta_0) = z_0$. Since $e^{\Delta_0} = 1 - z_0$, uniformly for $0 \leq \lambda \leq z_0$,

$$\frac{1}{1 - \gamma} \left[\left(\frac{1 - \lambda}{1 - z_0} \right)^{1-\gamma} - 1 \right] = z_0 - \lambda + O(z_0^2). \quad (48)$$

Integrating over $\lambda \in [0, z_0]$ gives

$$\mathcal{V}(\Delta_0) = \frac{\tau}{2\bar{\lambda}} z_0^2 + O(z_0^3) = O(z_0^2). \quad (49)$$

Consequently,

$$\Delta_0(m_0) = \frac{1}{\rho} \left[\ell_0 \pi(m_0) - \frac{\gamma}{2} \sigma_0^2 \right] + O(z_0^2). \quad (50)$$

Using $z_0 = 1 - e^{\Delta_0} = -\Delta_0 + O(\Delta_0^2)$ then gives

$$z_0(m_0) = \frac{1}{\rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m_0) \right] + O(z_0^2). \quad (51)$$

Since $G(z_0) = z_0/\bar{\lambda}$ in the interior-exit region,

$$x_0(m_0) = \frac{\tau}{\bar{\lambda}\rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m_0) \right] + O(z_0^2). \quad (52)$$

Dropping the second-order remainder and extending the expression across the no-exit margin gives

$$x_0^{\text{lin}}(m_0) := \frac{\tau}{\bar{\lambda}\rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m_0) \right]^+. \quad (53)$$

The linearized schedule in equation (53) is used only near the interior exit margin; it is not an approximation to the saturated-exit region. Combining equation (43), $\sigma_0 = \bar{\sigma} - \sigma$, and equation (52) proves Lemma 1. $\blacksquare$

B.3 Proof of Proposition 2.

Proof. Step 1: the exact value gap and monotonicity. Recall from Appendix B.2 that the exact fixed-capacity value gap satisfies

$$\rho \Delta_0(m) = \ell_0 \pi(m) - \frac{\gamma}{2} \sigma_0^2 + \mathcal{V}(\Delta_0(m)).$$

Wherever $\mathcal{V}$ is differentiable, the envelope theorem gives

$$\mathcal{V}_\Delta(\Delta) = -\frac{\tau}{\bar{\lambda}} \int_0^{\hat{z}(\Delta)} \left((1 - \lambda) e^{-\Delta} \right)^{1-\gamma} d\lambda \leq 0. \quad (54)$$

The same inequality holds in the one-sided sense at the kinks of the exit rule. Define

$$H(\Delta; m) := \rho \Delta - \mathcal{V}(\Delta) - \left[\ell_0 \pi(m) - \frac{\gamma}{2} \sigma_0^2 \right].$$

Because $\mathcal{V}$ is continuous and bounded,

$$H(\Delta; m) \rightarrow -\infty \quad \text{as } \Delta \rightarrow -\infty, \quad H(\Delta; m) \rightarrow \infty \quad \text{as } \Delta \rightarrow \infty.$$

Moreover,

$$H_\Delta(\Delta; m) = \rho - \mathcal{V}_\Delta(\Delta) \geq \rho > 0.$$

Thus, for every m , the exact value-gap equation has a unique solution $\Delta_0(m)$.

Since

$$\pi'(m) = -\omega'(m) = \frac{\bar{\omega}}{\bar{m} - \underline{m}} > 0,$$

implicit differentiation gives

$$\Delta'_0(m) = \frac{\ell_0 \pi'(m)}{\rho - \mathcal{V}_\Delta(\Delta_0(m))} > 0. \quad (55)$$

Therefore,

$$z'_0(m) = -e^{\Delta_0(m)} \Delta'_0(m) < 0,$$

and, since G is non-decreasing,

$$x'_0(m) = \tau G'(z_0(m)) z'_0(m) \leq 0$$

away from the kinks, with the corresponding one-sided inequality at the kinks. Lemma 1 then implies

$$\mu'_0(m) = \ell_0 \pi'(m) - x'_0(m) > 0.$$

Since $\sigma_0 = \bar{\sigma} - \sigma$ is independent of m ,

$$D'_0(m) = \frac{2\mu'_0(m)}{\sigma_0^2} > 0.$$

Hence D_0 is continuous and strictly increasing.

Step 2: existence and uniqueness of the exact threshold. Under

$$D_0(\underline{m}) < 1 < D_0(\bar{m}),$$

continuity and strict monotonicity imply that there exists a unique $m^\star \in (\underline{m}, \bar{m})$ satisfying $D_0(m^\star) = 1$. The classifications below, at, and above $m^\star$ follow immediately.

Step 3: the closed-form first-order threshold. Suppose the exact threshold lies in the small interior-exit region, and let $z_\star := z_0(m^\star)$. Lemma 1 gives

$$x_0(m^\star) = b \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m^\star) \right] + O(z_\star^2), \quad b := \frac{\tau}{\bar{\lambda} \rho}.$$

Thus,

$$\mu_0(m^\star) = (1 + b) \ell_0 \pi(m^\star) + (\gamma - 1) \sigma \sigma_0 - b \frac{\gamma}{2} \sigma_0^2 + O(z_\star^2). \quad (56)$$

At the exact threshold,

$$D_0(m^\star) = 1 \quad \Longleftrightarrow \quad \mu_0(m^\star) = \frac{1}{2} \sigma_0^2.$$

Consequently,

$$\pi(m^\star) = \frac{\frac{1}{2}\sigma_0^2(1+b\gamma) - (\gamma-1)\sigma\sigma_0}{(1+b)\ell_0} + O(z_\star^2). \quad (57)$$

Since

$$\pi(m) = \frac{\Delta a}{q_0} - \bar{\omega} \frac{\bar{m} - m}{\bar{m} - \underline{m}},$$

inverting equation (57) gives

$$m^\star = \bar{m}^\star + O(z_\star^2),$$

where $\bar{m}^\star$ is given in equation (22).

Step 4: recovery probability. For any fixed m , the auxiliary local wealth-share process is

$$\frac{d\eta_t}{\eta_t} = \mu_0(m) dt + \sigma_0 dZ_t.$$

Hence

$$\log \eta_t = \log \eta_0 + \left(\mu_0(m) - \frac{1}{2}\sigma_0^2 \right) t + \sigma_0 Z_t,$$

Its log drift is

$$\mu_0(m) - \frac{1}{2}\sigma_0^2 = \frac{1}{2}\sigma_0^2 (D_0(m) - 1).$$

For a fixed interior level $\bar{\eta} > \eta_0$, let

$$T_{\bar{\eta}} := \inf\{t \geq 0 : \eta_t \geq \bar{\eta}\}.$$

The standard hitting-probability formula for Brownian motion with drift gives

$$\Pr_{\eta_0}(T_{\bar{\eta}} < \infty) = \begin{cases} 1, & D_0(m) \geq 1, \\ \left(\frac{\eta_0}{\bar{\eta}}\right)^{1-D_0(m)}, & D_0(m) < 1. \end{cases} \quad (58)$$

When $m < m^\star$, $D_0(m) < 1$. Under the local wealth-share dynamics, $\log \eta_t$ converges to $-\infty$ almost surely, and the probability of reaching any fixed interior wealth share converges to zero as $\eta_0 \downarrow 0$. The wealth share nevertheless need not reach zero in finite time. At $m = m^\star$, $D_0(m) = 1$, so the log drift is zero. Recovery occurs almost surely, but its expected duration is infinite. When $m > m^\star$, the log drift is positive and recovery occurs almost surely in finite expected time. These results establish the boundary classification without relying on a stationary density.

Step 5: comparative statics of the first-order threshold. Define

$$T(b) := \frac{\frac{1}{2}\sigma_0^2(1+b\gamma) - (\gamma-1)\sigma\sigma_0}{(1+b)\ell_0}.$$

Then

$$\bar{m}^\star = \bar{m} - \frac{\bar{m} - \underline{m}}{\bar{\omega}} \left[\frac{\Delta a}{q_0} - T(b) \right].$$

Differentiating with respect to b gives

$$T'(b) = \frac{(\gamma - 1)\sigma_0 \left(\sigma + \frac{1}{2}\sigma_0 \right)}{\ell_0(1 + b)^2} > 0.$$

Since $\partial b / \partial \tau = 1 / (\bar{\lambda}\rho)$,

$$\frac{\partial \tilde{m}^\star}{\partial \tau} = \frac{\bar{m} - \underline{m}}{\bar{\omega}\bar{\lambda}\rho} \frac{(\gamma - 1)\sigma_0 \left(\sigma + \frac{1}{2}\sigma_0 \right)}{\ell_0(1 + b)^2} > 0.$$

Holding a_h , and hence q_0 , fixed,

$$\frac{\partial \tilde{m}^\star}{\partial \Delta a} = -\frac{\bar{m} - \underline{m}}{\bar{\omega}q_0} < 0.$$

Finally, using $\sigma_0 = \bar{\sigma} - \sigma$ and $\ell_0 = \bar{\sigma} / (\underline{\chi}\sigma)$,

$$\frac{\partial \tilde{m}^\star}{\partial \bar{\sigma}} = \frac{\bar{m} - \underline{m}}{\bar{\omega}} \frac{\underline{\chi}\sigma}{(1 + b)\bar{\sigma}^2} \left[(1 + b\gamma)\sigma_0 \left(\sigma + \frac{1}{2}\sigma_0 \right) - (\gamma - 1)\sigma^2 \right],$$

whose sign is generally ambiguous. ■

B.4 Proof of Corollary 3

Proof. The exact fixed-capacity value gap satisfies equation (47):

$$\rho\Delta_0(m) = \ell_0\pi(m) - \frac{\gamma}{2}\sigma_0^2 + \mathcal{V}(\Delta_0(m)).$$

Wherever the exit rule is differentiable, implicit differentiation gives

$$[\rho - \mathcal{V}_\Delta(\Delta_0(m))] \Delta'_0(m) = \ell_0\pi'(m) = -\ell_0\omega'(m). \quad (59)$$

From Appendix B.3,

$$\mathcal{V}_\Delta(\Delta) = -\frac{\tau}{\bar{\lambda}} \int_0^{\hat{z}(\Delta)} \left((1 - \lambda)e^{-\Delta} \right)^{1-\gamma} d\lambda \leq 0.$$

Hence the denominator in equation (59) is strictly positive and $\Delta'_0(m) > 0$.

Since

$$z_0(m) = 1 - e^{\Delta_0(m)}, \quad x_0(m) = \tau G(z_0(m)),$$

we have

$$z'_0(m) = -e^{\Delta_0(m)} \Delta'_0(m)$$

and

$$x'_0(m) = -\tau G'(z_0(m)) e^{\Delta_0(m)} \Delta'_0(m).$$

Differentiating the exact boundary drift,

$$\mu_0(m) = \ell_0\pi(m) + (\gamma - 1)\sigma\sigma_0 - x_0(m),$$

and substituting equation (59) gives

$$\begin{aligned}\mu'_0(m) &= -\ell_0\omega'(m) - x'_0(m) \\ &= -\ell_0\omega'(m) \left[1 + \frac{\tau G'(z_0(m)) e^{\Delta_0(m)}}{\rho - \mathcal{V}_\Delta(\Delta_0(m))} \right].\end{aligned}$$

Because $\omega'(m) < 0$, $G'(z) \geq 0$, and $\rho - \mathcal{V}_\Delta > 0$, the derivative is strictly positive. In the no-exit and saturated-exit regions, $G'(z_0) = 0$ away from the kinks, so the second term vanishes. At the kinks, the same monotonicity follows from the corresponding one-sided derivatives. ■

B.5 Proof of Proposition 3

Proof. Part (i). The derivation of the boundary value-gap equation in Appendix B.2 depends on the distribution of switching costs only through the exit-option term in the expert's boundary HJB. Replacing the uniform distribution with F_λ replaces $\mathcal{V}$ in equation (44) with $\mathcal{V}_F$ in equation (25), yielding equation (26). An expert exits if and only if

$$(1 - \lambda)\mathcal{J}_{h,0} > \mathcal{J}_{e,0},$$

or, equivalently, if $\lambda < \widehat{z}(\Delta_0)$. The realized exit hazard is therefore

$$x_0(m) = \tau F_\lambda(\widehat{z}(\Delta_0(m))).$$

Define

$$H(\Delta; m) := \rho\Delta - \mathcal{V}_F(\Delta) - \left[\ell_0\pi(m) - \frac{\gamma}{2}\sigma_0^2 \right].$$

On the interior-exit region, $\widehat{z}(\Delta) = 1 - e^\Delta$. Differentiating $\mathcal{V}_F$ with respect to Δ produces no boundary term because the integrand in equation (25) vanishes at $\lambda = \widehat{z}(\Delta)$. This is the value-matching condition for the marginal expert:

$$(1 - \lambda)e^{-\Delta} = 1.$$

Where $\widehat{z}$ is locally constant, the boundary term is also zero. It follows that

$$\mathcal{V}_{F,\Delta}(\Delta) = -\tau \int_0^{\widehat{z}(\Delta)} \left((1 - \lambda)e^{-\Delta} \right)^{1-\gamma} dF_\lambda(\lambda) \leq 0.$$

Moreover, $\mathcal{V}_F$ is C^1 , including at the kinks of $\widehat{z}$, where its one-sided derivatives coincide. Hence

$$H_\Delta(\Delta; m) = \rho - \mathcal{V}_{F,\Delta}(\Delta) \geq \rho > 0.$$

As $\Delta \rightarrow +\infty$, $\widehat{z}(\Delta) = 0$ and $\mathcal{V}_F(\Delta) = 0$, implying $H(\Delta; m) \rightarrow +\infty$. As $\Delta \rightarrow -\infty$, $\gamma > 1$ implies

$$\left((1 - \lambda)e^{-\Delta} \right)^{1-\gamma} \rightarrow 0.$$

Consequently, $\mathcal{V}_F(\Delta) \rightarrow \tau/(\gamma - 1)$, so $H(\Delta; m) \rightarrow -\infty$. Continuity and strict monotonicity

therefore imply that $H(\Delta; m) = 0$ has a unique solution $\Delta_0(m)$.

Part (ii). Because H is C^1 and $H_\Delta \geq \rho > 0$, the implicit function theorem gives

$$\Delta'_0(m) = \frac{\ell_0 \pi'(m)}{\rho - \mathcal{V}_{F,\Delta}(\Delta_0(m))} > 0,$$

where $\pi'(m) = -\omega'(m) > 0$. Since $\widehat{z}(\Delta)$ is non-increasing in Δ , the exit cutoff $\widehat{z}(\Delta_0(m))$ is non-increasing in capacity. The exit hazard

$$x_0(m) = \tau F_\lambda(\widehat{z}(\Delta_0(m)))$$

is therefore also non-increasing in m .

Part (iii). The boundary calculation in Lemma 1 depends on the switching-cost distribution only through the realized exit hazard. Hence

$$\mu_0(m) = \ell_0 \pi(m) + (\gamma - 1) \sigma \sigma_0 - x_0(m)$$

continues to hold. Because $\pi(m)$ is strictly increasing and $x_0(m)$ is non-increasing, $\mu_0(m)$ is strictly increasing. On the interior-exit region, $\widehat{z}(\Delta) = 1 - e^\Delta$. The chain rule and part (ii) imply

$$\begin{aligned} x'_0(m) &= -\tau f_\lambda(\widehat{z}(\Delta_0(m))) e^{\Delta_0(m)} \Delta'_0(m) \\ &= -\frac{\tau f_\lambda(\widehat{z}(\Delta_0(m))) e^{\Delta_0(m)}}{\rho - \mathcal{V}_{F,\Delta}(\Delta_0(m))} \ell_0 \pi'(m). \end{aligned}$$

Since $\widehat{z}(\Delta_0(m)) = z_0(m)$ on this region, substituting this expression into $\mu'_0(m) = \ell_0 \pi'(m) - x'_0(m)$ yields equation (27). In the interiors of the no-exit and saturated-exit regions, $\widehat{z}$ is locally constant. Thus $x'_0(m) = 0$ and

$$\mu'_0(m) = \ell_0 \pi'(m) = -\ell_0 \omega'(m) > 0.$$

The corresponding one-sided results hold at the kinks. Finally, $\sigma_0 = \bar{\sigma} - \sigma$ is independent of m , so

$$D_0(m) = \frac{2\mu_0(m)}{\sigma_0^2}$$

is continuous and strictly increasing.

Part (iv). D_0 is continuous and strictly increasing with $D_0(\underline{m}) < 1 < D_0(\bar{m})$, so the intermediate value theorem delivers a unique $m^\star \in (\underline{m}, \bar{m})$ with $D_0(m^\star) = 1$, and $D_0(m) < 1$ for $m < m^\star$, $D_0(m) > 1$ for $m > m^\star$. Definition 1 then classifies the boundary as locally trapping below $m^\star$ and locally escaping above it. ■

B.6 Proof of Corollary 4

Proof. In the small interior-exit region,

$$z_0 = 1 - e^{\Delta_0} \in (0, \bar{\lambda}), \quad \Delta_0 = \log(1 - z_0),$$

with z_0 close to zero. Uniformly for $\lambda \in [0, z_0]$,

$$\begin{aligned} \log \left[\left((1 - \lambda) e^{-\Delta_0} \right)^{1-\gamma} \right] &= (1 - \gamma) [\log(1 - \lambda) - \log(1 - z_0)] \\ &= (1 - \gamma)(z_0 - \lambda) + O(z_0^2). \end{aligned}$$

It follows that

$$\frac{((1 - \lambda) e^{-\Delta_0})^{1-\gamma} - 1}{1 - \gamma} = z_0 - \lambda + O(z_0^2).$$

Continuity of f_λ on its compact support implies $F_\lambda(z) = O(z)$ near zero. Therefore,

$$\begin{aligned} \mathcal{V}_F(\Delta_0) &= \tau \int_0^{z_0} (z_0 - \lambda) dF_\lambda(\lambda) + O(z_0^2) F_\lambda(z_0) \\ &= O(z_0^2), \end{aligned}$$

where the last equality follows from

$$\int_0^{z_0} (z_0 - \lambda) dF_\lambda(\lambda) \leq z_0 F_\lambda(z_0) = O(z_0^2).$$

Substituting this result into equation (26) and using $z_0 = -\Delta_0 + O(\Delta_0^2)$ gives

$$z_0(m) = \frac{1}{\rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m) \right] + O(z_0(m)^2).$$

Lipschitz continuity of f_λ at zero implies

$$F_\lambda(z) = f_\lambda(0)z + O(z^2).$$

Hence

$$\begin{aligned} x_0(m) &= \tau F_\lambda(z_0(m)) \\ &= \frac{\tau f_\lambda(0)}{\rho} \left[\frac{\gamma}{2} \sigma_0^2 - \ell_0 \pi(m) \right] + O(z_0(m)^2). \end{aligned}$$

Defining $b_F := \tau f_\lambda(0)/\rho$ yields the stated first-order hazard. The positive-part operator extends this expression across the no-exit margin, where $\widehat{z}(\Delta_0) = 0$ and $x_0 = 0$ exactly. At the capacity threshold, $D_0(m) = 1$, or equivalently $\mu_0(m) = \sigma_0^2/2$. Substituting the first-order exit hazard gives

$$(1 + b_F) \ell_0 \pi(m) = \frac{\sigma_0^2}{2} (1 + b_F \gamma) - (\gamma - 1) \sigma \sigma_0 + O(z_0(m)^2).$$

The first-order threshold therefore satisfies

$$\pi(\widetilde{m}_F^\star) = \frac{\frac{1}{2} \sigma_0^2 (1 + b_F \gamma) - (\gamma - 1) \sigma \sigma_0}{(1 + b_F) \ell_0}.$$

Since $\pi(m) = \Delta a/q_0 - \omega(m)$ and ω is strictly decreasing,

$$\tilde{m}_F^\star = \omega^{-1} \left(\frac{\Delta a}{q_0} - \frac{\frac{1}{2}\sigma_0^2(1 + b_F\gamma) - (\gamma - 1)\sigma\sigma_0}{(1 + b_F)\ell_0} \right).$$

Because ω' is continuous and strictly negative on $[\underline{m}, \bar{m}]$, it is bounded away from zero. Part (iii) of Proposition 3 then implies that $D'_0(m)$ is also bounded away from zero:

$$D'_0(m) \geq \frac{2\ell_0}{\sigma_0^2} |\omega'(m)|.$$

The error in the threshold is therefore of the same order as the terms omitted from the exit approximation:

$$m^\star = \tilde{m}_F^\star + O(z_\star^2).$$

Finally, if F_λ is uniform on $[0, \bar{\lambda}]$, then

$$f_\lambda(0) = \frac{1}{\bar{\lambda}}, \quad b_F = \frac{\tau}{\bar{\lambda}\rho} = b.$$

With the affine wedge in Assumption 1, the expression for $\tilde{m}_F^\star$ reduces to equation (22). ■

B.7 Proof of Proposition 4

Proof. Boundary capacity dynamics. Assumption 1 implies

$$\frac{\kappa}{\eta} \longrightarrow \ell_0, \quad \kappa = \ell_0\eta + o(\eta) \longrightarrow 0.$$

Because κ_t measures intermediary activity in equation (14), the rebuilding term that depends on current activity vanishes at the boundary. Capacity volatility also vanishes:

$$\sigma_m(\eta, m) = \nu\kappa^\beta(\bar{m} - m)(m - \underline{m}) \longrightarrow 0,$$

since $\beta > 1$. Substituting the exact exit schedule $x_0(m) = \tau G(z_0(m))$ into the remaining capacity drift yields

$$\dot{m} = g(m) := \kappa_m(\bar{m} - m) - \xi_m x_0(m)(m - \underline{m}), \quad \kappa_m := \kappa_m^1.$$

Equivalently, steady states are fixed points of

$$F(m) = \frac{\kappa_m \bar{m} + \xi_m x_0(m) m}{\kappa_m + \xi_m x_0(m)}.$$

Monotonicity of exit. Proposition 2 establishes that the boundary value gap $\Delta_0(m)$ is strictly increasing in capacity. It follows that

$$z'_0(m) = -e^{\Delta_0(m)} \Delta'_0(m) < 0.$$

Therefore, $x_0(m) = \tau G(z_0(m))$ is continuous and non-increasing.

Low-capacity steady state. Define

$$\widehat{m}_L = \frac{\kappa_m \bar{m} + \xi_m \tau \underline{m}}{\kappa_m + \xi_m \tau}.$$

The condition $z_0(\widehat{m}_L) > \bar{\lambda}$ implies that exit is saturated in a neighborhood of $\widehat{m}_L$, so $x_0(m) = \tau$ locally. Hence

$$g(\widehat{m}_L) = \kappa_m(\bar{m} - \widehat{m}_L) - \xi_m \tau(\widehat{m}_L - \underline{m}) = 0,$$

and

$$g'(\widehat{m}_L) = -(\kappa_m + \xi_m \tau) < 0.$$

Thus, $\widehat{m}_L$ is a locally asymptotically stable steady state.

Full-capacity steady state. The condition $z_0(\bar{m}) < 0$ implies that no exit occurs in a neighborhood of $\bar{m}$. Therefore,

$$x_0(\bar{m}) = 0, \quad g(\bar{m}) = 0, \quad g'(\bar{m}) = -\kappa_m < 0.$$

Interpreting stability relative to the state space $[\underline{m}, \bar{m}]$, $\bar{m}$ is locally asymptotically stable.

Interior steady states and basin separation. Because $g'(\widehat{m}_L) < 0$, there exists $\varepsilon_L > 0$ such that

$$g(m) < 0 \quad \text{for } m \in (\widehat{m}_L, \widehat{m}_L + \varepsilon_L).$$

Similarly, $g'(\bar{m}) < 0$ implies that there exists $\varepsilon_H > 0$ such that

$$g(m) > 0 \quad \text{for } m \in (\bar{m} - \varepsilon_H, \bar{m}).$$

Continuity of g therefore guarantees at least one interior steady state in $(\widehat{m}_L, \bar{m})$.

Suppose this interior steady state is unique and denote it by $\widehat{m}_U$. The preceding sign restrictions, together with the absence of any other zero, imply

$$g(m) < 0 \quad \text{for } m \in (\widehat{m}_L, \widehat{m}_U), \quad g(m) > 0 \quad \text{for } m \in (\widehat{m}_U, \bar{m}).$$

Trajectories starting below $\widehat{m}_U$ converge to $\widehat{m}_L$, whereas those starting above $\widehat{m}_U$ converge to $\bar{m}$. Thus, $\widehat{m}_U$ is unstable and separates the two basins of attraction. If $g'(\widehat{m}_U)$ exists and is nonzero, then $g'(\widehat{m}_U) > 0$.

Net-worth dynamics at the capacity steady states. From the definition of $D_0(m)$,

$$D_0(m) < 1 \quad \Longleftrightarrow \quad x_0(m) > \ell_0 \pi(m) + (\gamma - 1) \sigma \sigma_0 - \frac{1}{2} \sigma_0^2.$$

At the low-capacity steady state, $x_0(\widehat{m}_L) = \tau$. Hence

$$D_0(\widehat{m}_L) < 1 \quad \Longleftrightarrow \quad \tau > \bar{x}_0(\widehat{m}_L).$$

At full capacity, $x_0(\bar{m}) = 0$, so

$$D_0(\bar{m}) > 1 \quad \Longleftrightarrow \quad \bar{x}_0(\bar{m}) > 0.$$

Under these conditions, the expert-net-worth boundary is locally trapping at $\widehat{m}_L$ and locally escaping at $\bar{m}$. This classification describes the behavior of expert net worth at each capacity steady state; it is separate from the local stability of the capacity dynamics established above. ■

B.8 Proof of Proposition 1

Proof. When $\kappa_t < 1$, households operate a positive amount of capital, so their pricing condition holds with equality:

$$\mu_{R,h,t} - r_t = \epsilon_{h,t}.$$

The expert return advantage is therefore

$$\widetilde{\mu}_{R,e,t} - r_t = \epsilon_{h,t} + \frac{\Delta a}{q_t} - \omega(m_t).$$

In the benchmark economy, $\omega \equiv 0$ and $\tau = 0$.

Suppressing time subscripts and substituting the household pricing condition into equation (18) gives

$$\begin{aligned} \mu^\eta &= \frac{(1-\eta)\kappa}{\eta} \left[\epsilon_h + \frac{\Delta a}{q} - (1-\chi)\epsilon_h \right] - (1-\chi\kappa)\epsilon_h - (\ell_e - 1)v^2 \\ &= \frac{(1-\eta)\kappa}{\eta} \frac{\Delta a}{q} + (\ell_e - 1)\epsilon_h - (\ell_e - 1)v^2. \end{aligned}$$

The second equality uses

$$\frac{(1-\eta)\kappa}{\eta} \chi = \ell_e - \chi\kappa, \quad \ell_e := \frac{\chi\kappa}{\eta},$$

and hence

$$\frac{(1-\eta)\kappa}{\eta} \chi \epsilon_h - (1-\chi\kappa)\epsilon_h = (\ell_e - 1)\epsilon_h.$$

Equations (19)–(20) imply

$$\sigma^\eta = (\ell_e - 1)\sigma_R.$$

Applying Itô's lemma to $\log \eta$ therefore gives

$$\begin{aligned} \mu^B &= \mu^\eta - \frac{1}{2}(\ell_e - 1)^2 v^2 \\ &= \frac{(1-\eta)\kappa}{\eta} \frac{\Delta a}{q} + (\ell_e - 1)(\epsilon_h - v^2) - \frac{1}{2}(\ell_e - 1)^2 v^2, \end{aligned}$$

which is the first result.

The two volatility terms can be combined as

$$-(\ell_e - 1)v^2 - \frac{1}{2}(\ell_e - 1)^2 v^2 = -\frac{1}{2}(\ell_e - 1)(\ell_e + 1)v^2.$$

The leverage constraint in equation (13b) implies

$$\ell_e v \leq \bar{\sigma}, \quad v^2 \leq \frac{\bar{\sigma}^2}{\ell_e^2}.$$

It follows that

$$\mu^B \geq \underline{\mu}^B,$$

where the volatility term in $\underline{\mu}^B$ is

$$-\frac{\bar{\sigma}^2}{2} \left(1 - \frac{1}{\ell_e^2}\right) = -\bar{\sigma}^2 \left(1 - \frac{1}{\ell_e}\right) \left(\frac{1}{2} + \frac{1}{2\ell_e}\right).$$

The bound is attained when the leverage constraint binds. Because ϵ_h enters only through $(\ell_e - 1)\epsilon_h$,

$$\frac{\partial \underline{\mu}^B}{\partial \epsilon_h} = \ell_e - 1 > 0.$$

Under the constant-coefficient approximation, $\log \eta_s$ follows a Brownian motion with drift μ^B and volatility $(\ell_e - 1)v$. If $\underline{\mu}^B > 0$, the expected time required to reach $\eta_N > \eta_t$ is

$$\mathbb{E}_{\eta_t}[T_{\eta_N}] = \frac{\log(\eta_N/\eta_t)}{\mu^B} \leq \frac{\log(\eta_N/\eta_t)}{\underline{\mu}^B}.$$

The upper bound is strictly decreasing in ϵ_h .

In the full model, the intermediation wedge contributes

$$-\left[\frac{(1-\eta)\kappa}{\eta} + \kappa\right] \omega(m) = -\frac{\kappa}{\eta} \omega(m)$$

to the log wealth-share drift, while exit contributes $-\tau G(z_t)$. Thus,

$$\mu_t^F = \mu_t^B - \frac{\kappa_t}{\eta_t} \omega(m_t) - \tau G(z_t).$$

The derivatives with respect to $\epsilon_{h,t}$ hold $(\eta_t, \kappa_t, \ell_{e,t}, q_t, v_t)$ fixed. They are therefore conditional local derivatives, not general-equilibrium comparative statics. ■

C A microfoundation for the capacity process

This appendix derives the capacity process in equation (14) from the accumulation and destruction of relationship capital in the intermediary sector. A change of units maps relationship capital into intermediation capacity and gives the parameters of the capacity law their economic interpretation. The mapping from capacity to the intermediation wedge remains reduced form.

C.1 Relationship capital

Let $R_t \in [0, \bar{R}]$ denote the stock of match-specific relationship capital held by active experts. The ratio $R_t/\bar{R}$ measures the active share of potential borrower relationships. The potential mass is assumed to scale with the aggregate economy and is therefore normalized out. Relationship capital includes monitoring expertise, client information, and specialized personnel that support intermediation and reduce the cost of operating in the sector. In Section 4, I proxy for this stock using financial-firm franchise value and employment in credit intermediation. Infrastructure and public information that survive the loss of privately embodied relationships enter capacity through the floor $\underline{m}$ introduced below.

Relationship capital evolves through formation, destruction, and aggregate shocks.

Formation. New relationships are formed from the unutilized stock $\bar{R} - R_t$ at rate

$$\lambda_0 + \lambda_1 \kappa_t.$$

The parameter $\lambda_0 \geq 0$ captures baseline formation, while $\lambda_1 > 0$ captures learning by doing associated with the share κ_t of capital operated by experts.

Destruction through exit. Conditional on the aggregate state, experts face the same exit probability. Homotheticity makes the value gap common across experts, while exit opportunities and switching costs are independent across experts and independent of their holdings. The law of large numbers therefore implies that experts exiting during dt hold a fraction $x_t dt$ of aggregate relationship capital, where

$$x_t = \tau G(z_t)$$

is the exit hazard defined in Section 2.

Suppose each unit of relationship capital depends on its owner and on counterparty exposures with total effective weight $n \geq 0$. Owner exit destroys a fraction $d_0 \in [0, 1]$ of the unit. Counterparties exit independently at hazard x_t , generating an expected fractional loss $nd_1 x_t dt$, where $d_1 \in [0, 1]$. Consequently, the expected fractional loss over dt is

$$(d_0 + nd_1)x_t dt + o(dt).$$

Aggregate destruction is therefore $\zeta x_t R_t dt$, where

$$\zeta := d_0 + nd_1.$$

The parameter ζ is an effective measure of network exposure and may exceed one; it is not a count of counterparties.¹³ Destroyed relationships leave the stock R_t . There is no direct instantaneous cascade from the loss of one relationship to another. Instead, propagation occurs dynamically through

$$R \downarrow \implies \omega \uparrow \implies x \uparrow,$$

¹³Chodorow-Reich (2014) find that borrowers of impaired lenders obtain less credit on worse terms, Nguyen (2019) shows that lending remains depressed for up to six years after branch closures, and Huber (2018) documents persistent real effects of a lending contraction. Mitchener and Richardson (2019) document the propagation of lending contractions through interbank networks, building on the mechanism in Allen and Gale (2000). This evidence supports direct and network losses of relationship capital but does not identify the magnitude of ζ .

which is the feedback mechanism studied in Section 2.

Aggregate shocks. Sector-wide disturbances affect relationship capital through

$$\tilde{\nu} \kappa_t^\beta (\bar{R} - R_t) R_t dZ_t^m, \quad \tilde{\nu} > 0, \quad \beta > 1,$$

where Z_t^m has correlation φ with the capital shock Z_t^k . This volatility specification is imposed directly. Its product form makes capacity risk vanish at both bounds, while $\beta > 1$ ensures that it vanishes faster than intermediary activity as $\kappa_t \downarrow 0$, as required by Assumption 1. Combining the three components gives

$$dR_t = (\lambda_0 + \lambda_1 \kappa_t) (\bar{R} - R_t) dt - \zeta x_t R_t dt + \tilde{\nu} \kappa_t^\beta (\bar{R} - R_t) R_t dZ_t^m. \quad (60)$$

The interval $[0, \bar{R}]$ is invariant. The diffusion vanishes at both endpoints, while the drift satisfies

$$\mu_R|_{R=0} = (\lambda_0 + \lambda_1 \kappa_t) \bar{R} \geq 0, \quad \mu_R|_{R=\bar{R}} = -\zeta x_t \bar{R} \leq 0.$$

Thus, the drift points weakly inward at both boundaries and the diffusion cannot carry the process across either endpoint. A process initialized in $[0, \bar{R}]$ therefore remains in that interval.

C.2 Capacity

Define capacity as an affine transformation of relationship capital:

$$m_t := \underline{m} + (\bar{m} - \underline{m}) \frac{R_t}{\bar{R}} \in [\underline{m}, \bar{m}]. \quad (61)$$

The floor $\underline{m}$ captures intermediation inputs that are not embodied in private relationships. Because the transformation is affine, there is no Itô correction. Substituting equation (61) into equation (60) and defining

$$\kappa_m^1 := \lambda_0, \quad \kappa_m^2 := \lambda_1, \quad \xi_m := \zeta = d_0 + nd_1, \quad \nu := \tilde{\nu} \frac{\bar{R}}{\bar{m} - \underline{m}}, \quad (62)$$

yields the capacity process in equation (14), with $\sigma_m(\eta, m) = \nu \kappa^\beta (\bar{m} - m)(m - \underline{m})$. The transformation leaves the formation and destruction coefficients unchanged and rescales the volatility coefficient by $\bar{R}/(\bar{m} - \underline{m})$.

To make the normalization explicit, let $s_t := (m_t - \underline{m})/(\bar{m} - \underline{m}) = R_t/\bar{R}$. Its law of motion is

$$ds_t = (\lambda_0 + \lambda_1 \kappa_t)(1 - s_t) dt - \zeta x_t s_t dt + \nu (\bar{m} - \underline{m}) \kappa_t^\beta s_t (1 - s_t) dZ_t^m.$$

An affine rescaling of the capacity bounds leaves κ_m^1 , κ_m^2 , and ξ_m unchanged but rescales ν . When the wedge is expressed as a function of s_t , the normalized model therefore depends on capacity volatility through the product $\nu(\bar{m} - \underline{m})$, rather than on ν and the width of the capacity interval separately.

The relationship-capital construction provides a stock-flow foundation for the capacity process but does not derive the intermediation wedge. Relationship formation captures search, learning, and organizational accumulation, whose resource costs are not modeled separately. I interpret

$\omega(m)$ as the rent associated with specialized intermediation inputs, with $\omega'(m) < 0$. Assumption 1 imposes $\omega(m) = \bar{\omega}(1 - s)$, while Assumption 2 requires only that the wedge decrease with capacity. Because the wedge is rebated to households, it redistributes income rather than absorbing resources.

Two implications are worth emphasizing. First, capacity inherits its persistence from relationship capital. With exit shut down and expert activity fixed, the gap $\bar{R} - R_t$ closes at rate $\lambda_0 + \lambda_1 \kappa$. At full activity, the calibrated rate $\lambda_0 + \lambda_1 \approx 0.10$ per year implies a deterministic half-life of approximately seven years. This is the fastest recovery rate because relationship formation slows when expert activity declines. The deterministic half-life measures convergence to the capacity ceiling in the absence of exit and shocks. It differs from the autocorrelation half-life of stationary capacity reported in Table 2, which also reflects endogenous erosion and the bounded state space and is shorter in the calibrated model. The economy can leave the crisis region much earlier because recovery does not require capacity to return to its upper bound. The persistent decline in local credit following lender exit documented by [Nguyen \(2019\)](#) is consistent with this horizon.

Second, conditional on the exit-hazard path, the proportional erosion rate of the above-floor stock $m_t - \underline{m}$ is $\xi_m x_t$. Identifying ξ_m separately therefore requires independent information about exit. The calibrated ξ_m combines the direct loss of match-specific relationships with network spillovers. Holding formation and shocks fixed, an annualized exit hazard of one percentage point erodes the above-floor stock at the annual rate $0.01 \xi_m$, or ξ_m percent. The multiplier is not the number of relationships destroyed by each departing expert.

D Numerical solution

The equilibrium consists of two coupled second-order Hamilton–Jacobi–Bellman equations for the value multipliers $(\mathcal{J}_e, \mathcal{J}_h)$ on $(\eta, m) \in (0, 1) \times (\underline{m}, \bar{m})$. At each state, a static equilibrium maps the value multipliers and their derivatives into the capital price q , the allocations (κ, χ) , the price diffusion coefficients $(\sigma^{q,k}, \sigma^{q,m})$, and the risk-free rate r . I approximate the value multipliers using the neural-network PDE method of [Gopalakrishna and Wu \(2026\)](#). The algorithm alternates between solving the static equilibrium at each state and updating the two value functions.

Implementation. Each value multiplier is represented by a separate feed-forward network with inputs (η, m) , two hidden layers of 30 units, tanh activations, and a linear output. The networks are trained with Adam at a learning rate of 10^{-3} . Each outer iteration uses 1,500 gradient steps with mini-batches of 2,000 collocation points sampled from a $1,000 \times 40$ tensor grid on $[\underline{\eta}, 1 - \underline{\eta}] \times [\underline{m}, \bar{m}]$. I retain the network parameters associated with the lowest loss. The loss is the mean squared HJB residual augmented by a false-transient term that anchors the update to the preceding outer iteration.

At each grid point, the static block is solved to a residual tolerance of 10^{-6} , with at most 100 function evaluations in the unconstrained regime and 200 when the value-at-risk constraint binds. The constrained branch uses at most 12 bisection steps for the leverage cap and 8 damped Newton steps, with damping parameter 0.7 and tolerance 10^{-8} . The outer iteration stops when the mean absolute change in the value multipliers across the grid falls below 10^{-3} . The baseline solution converges in approximately twenty outer iterations.

Step 1: Static equilibrium. For given value multipliers and derivatives, I solve the static equilibrium separately at each grid point. When the VaR constraint is slack, the unknowns

$(\kappa, \sigma^{q,k}, \sigma^{q,m}, q)$ solve goods-market clearing, the expert pricing condition with $\Lambda^V = 0$, and the two price-amplification equations in (40) obtained by applying Itô's lemma to $q(\eta, m)$.

When the VaR constraint binds, the inside-equity requirement also binds, so $\chi = \underline{\chi}$. The same four unknowns then solve goods-market clearing, the two price-amplification equations, and

$$\frac{\chi^\kappa}{\eta} \|\sigma_R\| = \bar{\sigma}.$$

The multiplier on the VaR constraint is recovered from the expert pricing residual:

$$\Lambda^V = \tilde{\mu}_{R,e} - r - \underline{\chi} \epsilon_e - (1 - \underline{\chi}) \epsilon_h.$$

A constrained candidate is retained only if $\Lambda^V \geq 0$.

The physical-capital regime is selected using equation (16). When $\kappa < 1$, the household capital-pricing condition holds with equality; when $\kappa = 1$, the associated slack is weakly positive. At every grid point, I verify the VaR and household capital-allocation complementarity conditions, the bounds $0 \leq \kappa \leq 1$ and $\underline{\chi} \leq \chi \leq 1$, and all market-clearing conditions.

Step 2: Value-function update. Given the static equilibrium, I construct the HJB residuals $\mathcal{R}_e$ and $\mathcal{R}_h$ from equations (35)–(38) and update the two networks. The updated value multipliers are then returned to the static block, and the two steps are repeated until the outer convergence criterion is satisfied.

Boundary treatment. The capacity bounds are natural state constraints. The diffusion $\sigma_m = \nu \kappa^\beta (\bar{m} - m)(m - \underline{m})$ vanishes at both endpoints, and the drift points weakly into the state space. For the wealth share, I solve on $[\underline{\eta}, 1 - \underline{\eta}]$, using small numerical buffers around the degenerate endpoints $\eta = 0$ and $\eta = 1$.

Simulation. After convergence, I simulate (η_t, m_t) using Euler–Maruyama with a monthly time step, $dt = 1/12$. The wealth-share process uses the drift and both diffusion coefficients in equations (18)–(20). The innovations Z_t^k and Z_t^m have correlation φ , and capacity follows equation (14). I simulate 1,000 paths of 2,000 years each, discard the first 20% of every path as burn-in, and reflect Euler steps that cross the numerical state-space buffers. The moments in Table 2 are computed from the pooled post-burn-in sample. Crisis-state and event-time statistics use the definitions in Section 4.

E Additional figures for the benchmark model

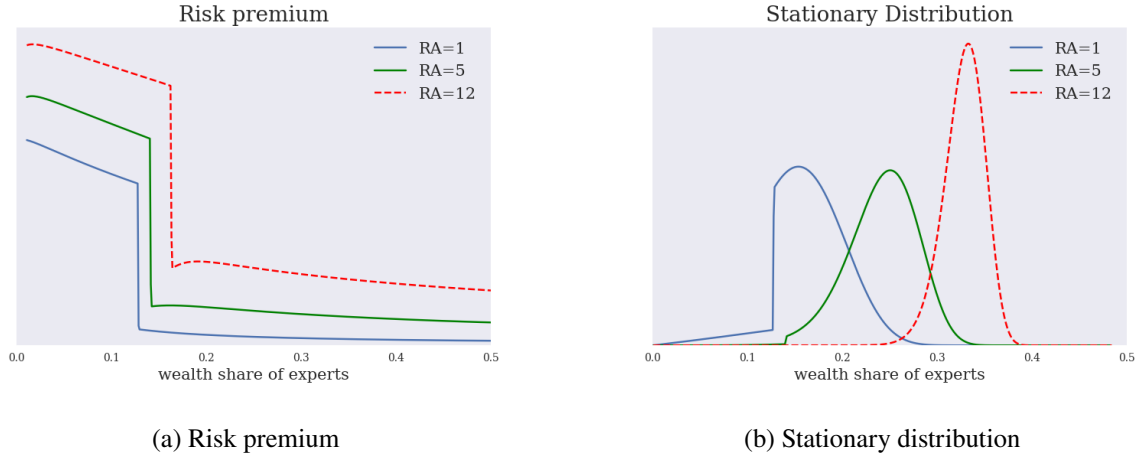


Figure 6. (a) The left panel presents the risk premium for $RA \in \{1, 5, 12\}$ as a function of the expert wealth share. (b) The right panel plots the stationary wealth-share distribution for the same three calibrations.

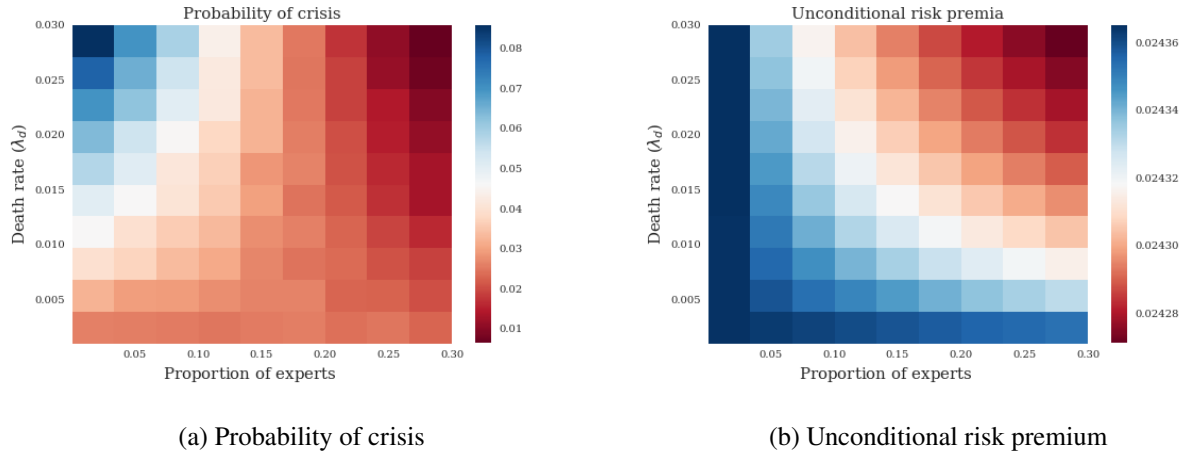


Figure 7. Heat maps across the OLG death rate λ_d (vertical axis) and the expert population share (horizontal axis). (a) Crisis probability. (b) Unconditional risk premium.